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Product index:

DRV/PMX/MV/001-D

ABCPMX SOFTWARE V1.20 COMPOSITE BEAMS DESIGN

VALIDATION REPORT

« Etablissement certifié qualité ISO 9001, le CTICM assure un suivi de
chaque étude dans le plus strict respect de ses procédures qualité »

Date :	15/05/2025	Auteur :	POM+GUD	Nombre total de pages : (y compris celle-ci)	116
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	Research and Valorisation Department		
	Project: ABCPMX-II Software - DAL-22-003		
	Product reference: DRV/PMX/MV/001	Rév. :	D

Révision	Date	Auteur	Objet
A	07/10/2024	GUD+POM	First release for V1.00
B	09/01/2025	POM	Introduction of new tests C06 and C07
C	15/05/2025	POM	Introduction of tests for fire design (F01 to F07, see chapter 7)
D	03/11/2025	POM	Introduction of tests for steel slim floor beams Correction of deflection of composite beams for Q loads
E			

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	Project: ABCPMX-II Software - DAL-22-003						
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	Date:	15/05/2025	Authors:	POM+GUD	Page:	1	Rév. : D

CONTENT

1	Object.....	4
2	References	4
2.1	First generation of Eurocodes	4
2.2	Drafts for the second generation of Eurocodes.....	4
2.3	Scientific references	4
3	Principles of validation	5
4	Scope grid	6
4.1	Normal situations.....	6
5	Tests cases – Composite beams.....	9
5.1	CASE C01: Single span composite beam.....	9
5.1.1	Description.....	9
5.1.2	Parameters	9
5.1.3	Verification of results.....	11
5.2	CASE C01b: Simply supported composite beam	14
5.2.1	Description.....	14
5.2.2	Parameters	14
5.2.3	Verification of results.....	14
5.3	CASE C02: Single span composite beam laterally restrained during its construction phase	15
5.3.1	Description.....	15
5.3.2	Parameters	15
5.3.3	Verification of results.....	16
5.4	CASE C03: Single span composite beam with partial encasement (construction phase).....	17
5.4.1	Description.....	17
5.4.2	Parameters	17
5.4.3	Verification of results.....	20
5.5	CASE C04: Single span composite beam with partial encasement (composite phase)	23
5.5.1	Description.....	23
5.5.2	Parameters	23
5.5.3	Verification of results.....	24
5.6	CASE C05: Composite beam with cantilever span.....	29
5.6.1	Description.....	29
5.6.2	Parameters	29
5.6.3	Verification of results.....	31
5.7	CASE C05b: Composite beam with cantilever span.....	35
5.7.1	Description.....	35
5.7.2	Parameters	35
5.7.3	Verification of results.....	36
5.8	CASE C06: Composite beam with a solid slab.....	38
5.8.1	Description.....	38
5.8.2	Parameters	38
5.8.3	Verification of results.....	40

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	2	Rév. : D

5.9	CASE C07: Composite beam with a composite slab and Cofraplus 220	42
5.9.1	Description.....	42
5.9.2	Parameters	42
5.9.3	Verification of results.....	44
6	Tests cases – STEEL beams.....	47
6.1	CASE S01: Steel beam with 2 cantilever spans.....	47
6.1.1	Description.....	47
6.1.2	Parameters	47
6.1.3	Verification of results.....	49
6.2	CASE S02: Steel beam with welded profile	52
6.2.1	Description.....	52
6.2.2	Parameters	52
6.2.3	Verification of results.....	54
6.3	CASE S02b: Steel beam with welded profile	56
6.3.1	Description.....	56
6.3.2	Parameters	56
6.3.3	Verification of results.....	57
6.4	CASE S03: Steel beam with welded profile and MV interaction	59
6.4.1	Description.....	59
6.4.2	Parameters	59
6.4.3	Verification of results.....	61
6.5	CASE S03b: Steel beam with welded profile and MV interaction.....	63
6.5.1	Description.....	63
6.5.2	Parameters	63
6.5.3	Verification of results.....	63
6.6	CASE S04: Steel beam with welded profile and interaction between bending and shear buckling	64
6.6.1	Description.....	64
6.6.2	Parameters	64
6.6.3	Verification of results.....	66
6.7	CASE S04b: Steel beam with welded profile and interaction between bending and shear buckling.....	68
6.7.1	Description.....	68
6.7.2	Parameters	68
6.7.3	Verification of results.....	68
7	Tests cases – Fire design.....	70
7.1	CASE F01: Steel beam with unprotected hot rolled section	70
7.1.1	Description.....	70
7.1.2	Parameters	70
7.1.3	Verification of results.....	71
7.2	CASE F01b: Steel beam with unprotected galvanized hot rolled section	72
7.2.1	Description.....	72
7.2.2	Parameters	72
7.2.3	Verification of results.....	72
7.3	CASE F02: Steel beam with protected welded section (spray).....	73

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	3	Rév. : D

7.3.1	Description.....	73
7.3.2	Parameters	73
7.3.3	Verification of results.....	74
7.4	CASE F02b: Steel beam with protected welded section (intumescent paint).....	75
7.4.1	Description.....	75
7.4.2	Parameters	75
7.4.3	Verification of results.....	75
7.5	CASE F02c: Steel beam with protected welded section (plaster boards)	77
7.5.1	Description.....	77
7.5.2	Parameters	77
7.5.3	Verification of results.....	77
7.6	CASE F03: Composite beam with unprotected hot rolled profile	79
7.6.1	Description.....	79
7.6.2	Parameters	79
7.6.3	Verification of results.....	80
7.7	CASE F03b: Composite beam with galvanised hot rolled profile	82
7.7.1	Description.....	82
7.7.2	Parameters	82
7.7.3	Verification of results.....	82
7.8	CASE F04: Composite beam with protected welded profile	84
7.8.1	Description.....	84
7.8.2	Parameters	84
7.8.3	Verification of results.....	86
7.9	CASE F04b: Composite beam with protected welded profile (board).....	88
7.9.1	Description.....	88
7.9.2	Parameters	88
7.9.3	Verification of results.....	89
7.10	CASE F05: Composite beam with protected welded profile and solid slab.....	91
7.10.1	Description.....	91
7.10.2	Parameters	91
7.10.3	Verification of results.....	92
7.11	CASE F05B: Composite beam with protected welded profile and solid slab.....	92
7.11.1	Parameters	92
7.11.2	Verification of results.....	93
7.12	CASE F06: Composite beam partially encased profile	94
7.12.1	Description.....	94
7.12.2	Parameters	94
7.12.3	Verification of results.....	95
7.13	CASE F07: Composite beam with protected profile and unfilled voids.....	96
7.13.1	Description.....	96
7.13.2	Parameters	96
7.13.3	Verification of results.....	98

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	4	Rév. : D

1 OBJECT

ABC-PMX software aims at the calculation of composite beams according to EN 1994-1-1. Its development is co-funded by ArcelorMittal and CTICM. This document constitutes the Validation Manual.

2 REFERENCES

2.1 First generation of Eurocodes

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- [3] EN 1992-1-1 – Design of concrete structures. Part 1-1: General rules and rules for buildings. AFNOR. October 2005.
- [4] EN 1993-1-1 – Design of steel structures. Part 1-1: General rules and rules for buildings. AFNOR. October 2005.
- [5] EN 1993-1-2 – Design of steel structures. Part 1-2: General rules - Structural fire design
- [6] EN 1993-1-5 – Design of steel structures. Part 1-5: Plated structural elements. AFNOR. March 2007.
- [7] EN 1994-1-1 – Design of composite steel and concrete structures. Part 1-1: General rules and rules for buildings. AFNOR. June 2005.
- [8] EN 1994-1-2 – Design of composite steel and concrete structures. Part 1-2: General rules - Structural fire design

2.2 Drafts for the second generation of Eurocodes

- [9] prEN 1992-1-1 – Design of concrete structures. Part 1-1: General rules and rules for buildings. Document SC2N2034
- [10] FprEN 1993-1-1 – Design of steel structures. Part 1-1: General rules and rules for buildings. Document SC3 N3504 – 26/11/21
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- [12] prEN 1994-1-1 – Design of composite steel and concrete structures. Part 1-1: General rules and rules for buildings. CEN/TC250/SC4 document N 2394. November 2021 - Draft.

2.3 Scientific references

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- [17] Déversement des poutres en acier – Collection CTICM (in French)
- [18] Guide de conception pour les planchers minces à poutres incorporées - ECCS European Convention for Constructional Steelwork – Report no 83 – 1996
- [19] COSFB Software – Technical specifications – 15/09/2020 – ArcelorMittal

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	5	Rév. : D

[20] Nomogrammes – Méthode graphique pour déterminer la résistance au feu des structures en acier, selon l'EN 1993-1-2 : 2005 – Centre Information Acier

3 PRINCIPLES OF VALIDATION

This report includes test cases for all configurations. For each of them, the following are specified:

- the input data of the test case ;
- software output data ;
- the results obtained by the reference ;
- the comparison between the software output results and those obtained by the reference.

The input data is strictly identical between the software and the reference.

The references values used for software validation come from Excel spreadsheets developed specifically at CTICM and scientific literature references.

The comparison of the results is based on the evaluation of the differences between the values provided by the software and those of the reference. This comparison will be limited to the main results (verification ratios). The intermediate results will therefore be neither detailed nor compared here.

The analysis of the results obtained and the deviations observed make it possible to confirm that the ABC-PMX-II software produces values in accordance with the calculation methods applied.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	6	Rév. : D

4 SCOPE GRID

4.1 Normal situations

Parameters	Test C01	Test C01b	Test C02	Test C03	Test C04
Type of section / beam	Composite beam	Composite beam	Composite beam	Composite beam / partial encasement	Composite beam / partial encasement
Type of profile	HR	HR	HR	HR	HR
Steel grade	S275 (EC3)	S275 (EC3)	S275 (EC3)	S355 M/ML	S355 EC3
Construction phase	No	yes	Yes	Yes	No
Propping	unpropped	unpropped	unpropped	unpropped	unpropped
Cantilever spans	No	No	No	No	No
Type of slab	Composite	Composite	Composite	Composite	Composite
Steel deck label	Cofraplus 60	Cofraplus 60	Cofraplus 60	Cofraplus 60	Cofraplus 60
Ribs orientations	Perpendicular	Perpendicular	Perpendicular	Perpendicular	Perpendicular
Restraint by the deck	-	-	Yes	No	No
Studs	19-100	19-100	-	-	19-100
Standard	Generation 1	Generation 1	Generation 1	Generation 1	Generation 1
Type of design	Plastic	Plastic			

Parameters	Test C05	Test C05b	Test C06	Test C07	
Type of section / beam	Composite beam	Composite beam	Composite beam	Composite beam	
Type of profile	HR	HR	HR	HR	
Steel grade	S355 JR	S355 JR	S460	S420	
Construction phase	No	No	No	No	
Propping	unpropped	unpropped	propped	propped	
Cantilever spans	left	right	no	no	
Type of slab	Composite	Composite	Solid	Composite	
Steel deck label	Cofraplus 60	Cofraplus 60	/	Cofraplus 220	
Ribs orientations	Perpendicular	Perpendicular	/	Perpendicular	
Restraint by the deck	-	-	/	-	
Studs	19-100	19-100	22-125	22-100	
Standard	Generation 1	Generation 1	Generation 1	Generation 1	
Type of design	Plastic	Plastic	Plastic	Plastic	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	7	Rév. : D

Parameter	Test S01	Test S02	Tests S03, S03b	Tests S04, S04b	
Type of section / beam	Steel beam	Steel beam	Steel beam	Steel beam	
Type of profile	HR	Welded	Welded unsymmetrical	Welded unsymmetrical	
Steel grade	S355 M/ML	S355 JR	S275 JR	S460 EC3	
Construction phase	-	-	-	-	
Propping	-	-	-	-	
Cantilever spans	Left and right	No	Left or right	Left	
Type of slab	Solid	Solid	Solid	Solid	
Steel deck label	-	-	-	-	
Ribs orientations	-	-	-	-	
Restraint by the deck	-	-	-	-	
Studs	-	-	-	-	
Standard	Generation 1	Generation 1	Generation 1	Generation 1	
Type of design	Plastic	Plastic	Plastic	Elastic	
Shear buckling	No	Yes	No	Yes	
Interaction MV	No	No	Yes	Yes	
Interaction M shear buckling	No	No	No	Yes	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	8	Rév. : D

Parameters	Test F05b	Test F06
Type of section / beam	Composite beam	Composite beam partially encased
Type of profile	Welded	HR
Fire protection	Spray	Partial encasement
Steel grade	S355	S355
Cantilever spans	No	No
Type of slab	Solid	Composite
Steel deck label		Cofraplus 60
Ribs orientations		Perpendicular
Studs	22x125	19x100
Standard	Generation 1	Generation 1
Heating of the slab	Numerical	Standard

5 TESTS CASES – COMPOSITE BEAMS

5.1 CASE C01: Single span composite beam

5.1.1 Description

This test is based on the *Construction Métallique* review n°2018-2. It deals with a single span composite beam with a composite slab.

The reference values are taken from the paper. Because the calculation method of the software is more accurate, some values of the review may not match perfectly.

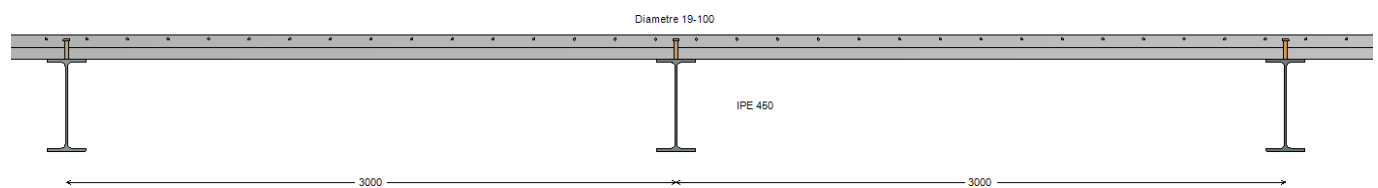
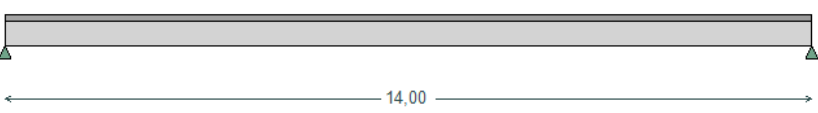
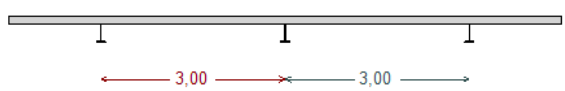


Figure 1 : Transversal view

5.1.2 Parameters

Definition of the beam:

Span lengths Main span L = 14 m ☐ Left cantilever span ☐ The slab is continuous at the left support ☐ Right cantilever span ☐ The slab is continuous at the right support		
Spacings ☒ Intermediate beam ☐ Edge beam Spacings $d_1 = 3$ m $d_2 = 3$ m	Slab openings ☐ Left opening ☐ Right opening	

Definition of the slab:

Composite slab

Total depth: $t_d = 120$ mm

Perpendicular ribs

Steel profiled sheeting: Cofraplus 60, thickness 0,75 mm

Prepunched sheeting

Definition of the profile:

Hot rolled profile

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	10	Rév. : D

Profile: IPE 450
Steel grade: S275 EC3

Definition of the concrete:

Normal concrete
Concrete class: C25/30
Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the connection:

Studs: 19x100
One stud in each rib
Unpropped beam during construction

Definition of loads:

G1: self-weight only
G2: surface load $1,4 \text{ kN/m}^2$
live load Q1: surface load $2,5 \text{ kN/m}^2$
 $\psi_2 = 0,3$

Combination of loads:

ULS: $1,35 (G1+G2) + 1,5 Q1$
SLS: $(G1+G2) + Q1$

Design parameters:

Standard: Eurocodes first generation
Design: According to the class of cross-section
Relative humidity: 50 %
Shrinkage deformation: 325×10^{-6}
Age of concrete: $t_0 = 28 d$ for G2
 $t = 50 \times 365 d$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	11	Rév. : D

5.1.3 Verification of results

CASE C01 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	98.8	98.8	
Shear area : $A_v =$ (cm ²)	50.9	50.8	0.20%
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	33 740	33 743	0.01%
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 676	1 672	0.24%
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 500	1 500	
Elastic section modulus (z-z axis) : $W_{el,z} =$ (cm ³)	176	176	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	1 702	1 702	
Plastic section modulus (z-z axis) : $W_{pl,z} =$ (cm ³)	276	272.8	1.17%
Torsional constant : $I_t =$ (cm ⁴)	66.66	66.75	0.13%
Warping constant : $I_w =$ (cm ⁶)	780 739	791 005	1.30%
Mass by length unit : $m_{lin} =$ (kg/ml)	77.6	77.6	
Total mass : $m =$ (kg)	1086	1 086	
Total paint surface : $S =$ (cm ²)	224 700	224 721	0.01%
Paint surface (without upper face) : $S =$ (cm ²)	198 100	198 121	0.01%
Massiveness : $M =$ (m ⁻¹)	162	162.4	0.25%
Massiveness (without upper face) : $M =$ (m ⁻¹)	143	143.2	0.14%
Nominal yield strenght : $f_y =$ (MPa)	275	275	
Young's modulus : $E =$ (MPa)	210 000	210 000	
Density : $\rho =$ (kg/m ³)	7 850	7 850	
Tensile resistance : $N_{t,Rd} =$ (kN)	2 717	2 718	0.04%
Shear resistance : $V_{pl,Rd} =$ (kN)	807	807	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	468	468	
Elastic moment resistant : $M_{el,Rd} =$ (kN.m)	412.5	412	0.12%
Section class for bending :	1	1	
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	44,8	44,8	
Shear buckling resistance : $72\varepsilon/\eta =$	66,5	66,5	
Verification of web buckling required?	No	No	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	12	Rév. : D

CASE C01 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	124	124	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	
Sec. moment of area per unit of width : $I_{pu} =$ (cm ⁴ /m)	44.37	44.37	
Shear resistance of the connector : $P_{Rd} =$ (kN)	52.9	52.9	
Reduction factor : $k_t =$	0.71	0.71	

CASE C01 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Effective width of the slab : $b_{eff} =$ (m)	3,0	3,0	
Modular ratio : $n(G_2) =$	27.5	27.5	
Modular ratio : $n(Q_1) =$	6.67	6.67	
Modular ratio : $n(SHC) =$	26.1	26.1	
Second moment of area (y-y axis) : $I_{el,y}(n = 6.67)$ (cm ⁴)	106 570	106 571	
Second moment of area (y-y axis) : $I_{el,y}(n = 26.1)$ (cm ⁴)	74 820	74 821	
Second moment of area (y-y axis) : $I_{el,y}(n = 27.5)$ (cm ⁴)	73 533	73 534	
Second moment of area (y-y axis) : $I_{el,y}(steel)$ (cm ⁴)	33 742	33 743	
Elastic neutral axis : $z_{el,y}(n = 6.67)$ (mm)	6.83	6.82	0.15%
Elastic neutral axis : $z_{el,y}(n = 26.1)$ (mm)	-93.4	-93.4	
Elastic neutral axis : $z_{el,y}(n = 27.5)$ (mm)	-97.5	-97.5	
Elastic neutral axis : $z_{el,y}(steel)$ (mm)	-225	-225	
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	846	846	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	13	Rév. : D

CASE C01 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	50.8	50.8	
Bending moment : $M(G_1) =$ (kN.m)	178	178	
Shear force : $V(G_2) =$ (kN)	29.4	29.4	
Bending moment : $M(G_2) =$ (kN.m)	103	103	
Shear force : $V(Q_1) =$ (kN)	52.5	52.5	
Bending moment : $M(Q_1) =$ (kN.m)	184	184	
Shear force : $V(Q_c) =$ (kN)	15	15	
Bending moment : $M(Q_c) =$ (kN.m)	64.9	64.9	

CASE C01 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.84	0.84	
Resistance to shear force : $\Gamma_V =$	0.23	0.23	
Resistance to MV interaction : $\Gamma_{MV} =$	0.84	0.84	
Degree of shear connection : $\eta =$	0.68	0.68	
Minimum degree of connection : $\eta_{\min} =$	0.57	0.57	
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	2.06	2.06	
Min. compression strut inclination angle : $\theta_{f,\min} =$ (°)	27	27	
Resistance of the compression strut : $\Gamma_{sf} =$	0.57	0.57	
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	0	0	

CASE C01 RESULTS: ULS AT CONSTRUCTION STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.72	0.72	
Resistance to shear force : $\Gamma_V =$	0.11	0.11	

CASE C01 RESULTS: SLS ASSUMING FULL SHEAR CONNECTION	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	51.2	51.2	
Deflection : $\delta(G_2)$ (mm)	13.6	13.6	
Deflection : $\delta(Q_1)$ short term only (mm)	16.8	16.8	
Deflection : $\delta(Q_1)$ including ψ_2 fraction as long term effect (mm)	18,9	19,0	0,4 %
Deflection : $\delta(SHC)$ (mm)	13.8	13.9	0.72%
Deflection : $\delta(SLS)$ (Q1 short term only) (mm)	95.5	95.5	
Deflection : $\delta(SLS)$ (Q1 including ψ_2 fraction as long term effect) (mm)	97,7	97,7	
Eigen frequency : $f(G + 0.2Q_1)$ (Hz)	3.3	3.3	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	14	Rév. : D

5.2 CASE C01b: Simply supported composite beam

5.2.1 Description

This test is based on the *Construction Métallique* review n°2018-2. It deals with a single span composite beam with a composite slab in the construction phase.

It corresponds to the construction phase of the Case C01.

5.2.2 Parameters

All parameters are the same as in Case C01, with the additional parameters:

Lateral restraint of the profile:

No lateral restraints in the span

No restraint by the steel profiled sheeting

Definition of loads:

G1: self-weight only

live load QC: surface load 0,5 kN/m²

distributed load : 3 kN/m on the 3 m at the center of the span

Combination of loads:

ULS: 1,35 G1 + 1,5 QC

5.2.3 Verification of results

CASE C01b RESULTS: ULS AT CONSTRUCTION STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.72	0.72	
Resistance to shear force : $\Gamma_V =$	0.11	0.11	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	15	Rév. : D

5.3 CASE C02: Single span composite beam laterally restrained during its construction phase

5.3.1 Description

This test is based on the *Construction Métallique* review n°2022-2. It deals with a single span composite beam which is laterally restrained by the steel deck during its construction phase.

The reference values are taken from the paper. Because the calculation method of the software is more accurate, some values of the review may not match perfectly.

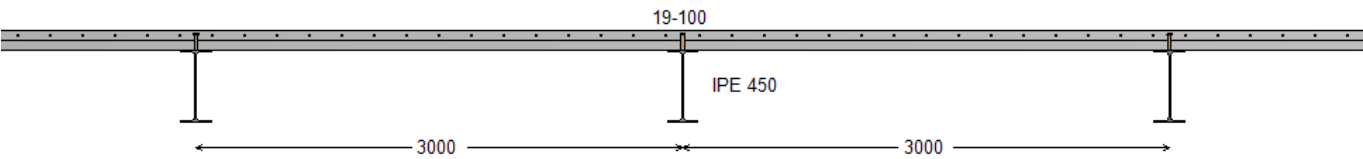


Figure 2 : Transversal view

5.3.2 Parameters

All parameters are the same as in Case C01, with the additional parameters:

Lateral restraint of the profile:

- No lateral restraints in the span
- Restraint by the steel profiled sheeting:

Paramètres du sol ☒ Poutre est maintenue latéralement par le bac Nombre de tôles dans la direction transversale: 2 Dimensions du sol (travée principale) Longueur de travée: 14,00 m Largeur: 12,00 m Type de transition entre deux tôles adjacentes: Tôles adjacentes sans jeu	Fixation du bac à la poutre Une attache: à chaque onde Type d'attaches: Fixations par pistolet à cartouches Diamètre: De 3,7 à 4,8 mm Glissement: 0,10 mm/kN	Paramètres Rigidité au cisaillement K = 0,176 α_5 = 0,500 c ₁₁ = 0,824 mm/kN c ₁₂ = 0,120 mm/kN c ₂₁ = 0,064 mm/kN c ₂₂ = 0,064 mm/kN c = 1,073 mm i _{act} = 13049 kN/m Rigidité à la flexion k _{theta,A} = 18,2 kN.m/m k _{theta,C} = 186 kN.m/m k _{theta} = 16,5 kN.m/m ☒ Prise en compte de la rigidité flexionnelle du bac
Tôle individuelle Nombre de travée traversée par les tôles: 2 Dimensions Longueur: 6,00 m Largeur: 1,04 m	Couturage Type d'attaches: Vis autoforeuses Diamètre: De 4,1 à 4,8 mm Glissement: 0,25 mm/kN Espacement: 500 mm	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	16	Rév. : D

Definition of loads:

G1: self-weight only
live load QC: surface load 0,5 kN/m²
distributed load : 3 kN/m on the 3 m at the center of the span

Combination of loads:

ULS: 1,35 G1 + 1,5 QC

5.3.3 Verification of results

CASE C02 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	50.8	50.8	
Bending moment : $M(G_1) =$ (kN.m)	178	178	
Shear force : $V(Q_c) =$ (kN)	15	15	
Bending moment : $M(Q_c) =$ (kN.m)	64.8	64.9	
Shear force : $V_{Ed} =$ (kN)	91	91	
Bending moment : $M_{Ed} =$ (kN.m)	337	337	

CASE C02 RESULTS: ULS during construction stage	Reference	ABCPMX	Dev.
Critical moment to lateral torsional buckling (without flexural bending of deck accounted for) : M_{cr} (kN.m)	2 260	2 304	-1.91%
Critical moment to lateral torsional buckling (with flexural bending of deck accounted for) : M_{cr} (kN.m)	3 237	3 299	-1.88%
Resistance to lateral torsional buckling (with flexural bending of deck accounted for) : $\Gamma_{LT} =$	0.77	0.77	

5.4 CASE C03: Single span composite beam with partial encasement (construction phase)

5.4.1 Description

This test is based on the paper in the *Construction Métallique* review n°2023-3. It deals with a single span composite beam with a composite slab and partial encasement during its construction phase.

The values from the paper are used as references. Because the calculation method of the software is more accurate, some values of the review may not match perfectly.

It corresponds to the construction phase of the case C04.

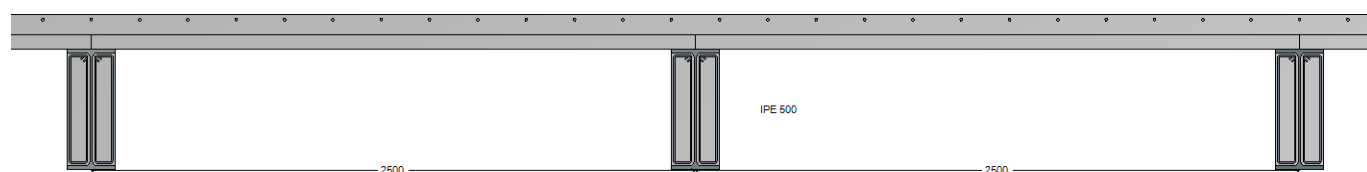
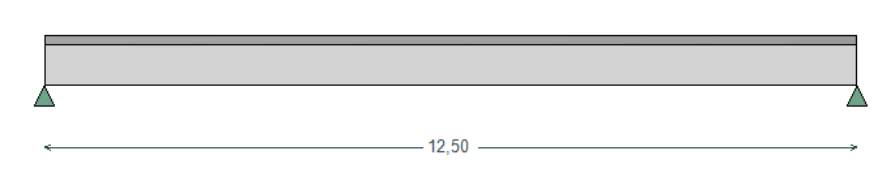
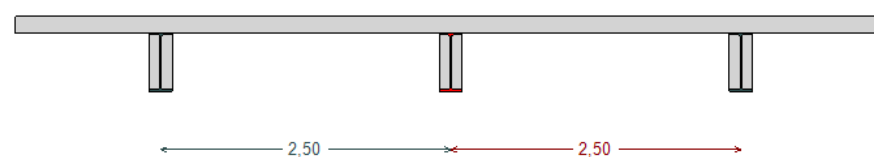


Figure 3 : Transversal view

5.4.2 Parameters

Definition of the beam:

Span lengths Main span L = 12,50 m ☐ Left cantilever span ☐ Right cantilever span	
Spacings ☒ Intermediate beam ☐ Edge beam Spacings $d_1 = 2,5$ m $d_2 = 2,5$ m	

Definition of the slab:

Composite slab

Total depth: $t_d = 140$ mm

Perpendicular ribs

Continuous ribs with un-continuous deck

Steel profiled sheeting: Cofraplus 60, thickness 1 mm

Prepunched sheeting

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	18	Rév. : D

Unpropped beam during construction

Definition of the profile:

Hot rolled profile
Profile: IPE A 500
Steel grade: S355 M/ML

Definition of the concrete:

Normal concrete
Concrete class: C25/30
Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the partial encasement:

Dimensions

Width

$b_c = 1$

x

b_y

$b_c = 200$

mm

Stirrups

Type

Closed stirrups

Diameter

$\phi_e = 6$

mm

Coverage

$u_y = 10$

mm

$u_z = 48$

mm

Concrete

Class

C25/30

$f_{ck} = 25$

MPa

$E_{cm} = 31,476$

GPa

Steel of the reinforcement

Class

B500

$f_{sk} = 500$

MPa

Longitudinal reinforcement

External

Middle

Internal

Number

Single

None

Single

Diameter

8

8

12

mm

Active rebars

☒

☐

☒

Position

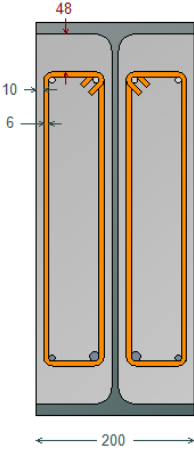
$z_{s,inf} = 423$

mm

Lower layer

$A_{s,inf} = 163,363$

mm²



	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	19	Rév. : D

Lateral restraint of the profile:

No lateral restraints in the span

No restraint by the steel profiled sheeting

Definition of loads:

G1: self-weight only

live load QC: distributed load: 1,86 kN/m on the full length

distributed load: 1,05 kN/m on the 3 m at the centre of the span

Combination of loads:

ULS: 1,35 G1 + 1,5 QC

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	20	Rév. : D

5.4.3 Verification of results

CASE C03 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	101	101.1	
Shear area : $A_v =$ (cm ²)	50.4	50.4	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	42 930	42 933	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 949	1 936	0.18%
Elastic section modulus about y-y axis : $W_{el,y} =$ (cm ³)	1 728	1 728	
Elastic section modulus about z-z axis : $W_{el,z} =$ (cm ³)	193,9	193.6	0.18%
Plastic section modulus about y-y axis : $W_{pl,y} =$ (cm ³)	1 946	1 946	
Plastic section modulus about z-z axis : $W_{pl,z} =$ (cm ³)	301.6	298.3	1.15%
Torsional moment : $I_t =$ (cm ⁴)	64.18	64.28	-0.16%
Warping moment : $I_w =$ (cm ⁶)	1 125 000	1 125 230	
Mass by length unit : $m_{lin} =$ (kg/ml)	79,4	79,4	
Mass : $m =$ (kg)	994	992	0.16%
Total paint surface : $S =$ (cm ²)	217 643	217 643	
Paint surface (without upper face) : $S =$ (cm ²)	192 643	192 643	
Section factor : $M =$ (m ⁻¹)	172.0	172.2	-0.14%
Section factor (without upper face) : $M =$ (m ⁻¹)	152.2	152.4	-0.13%
Nominal yield strength : $f_y =$ (MPa)	355	355	
Axial resistance : $N_{t,Rd} =$ (kN)	3 595	3 589	0.15%
Shear resistance : $V_{pl,Rd} =$ (kN)	1 036	1 033	0.32%
Plastic resistance to bending : $M_{pl,Rd} =$ (kN.m)	692	691	0.16%
Elastic resistance to bending : $M_{el,Rd} =$ (kN.m)	614	613	0.24%
Section class for pure bending :	1	1	
Section class for pure compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	50.7	50.7	
Shear buckling resistance : $124\varepsilon =$	101	101	
Verification of web buckling required?	No	No	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	21	Rév. : D

CASE C03 RESULTS: PARTIAL ENCASEMENT	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	191	191	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	

CASE C03 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	210	210	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	
Sec. moment of area per unit of width : $I_{pu} =$ (cm ⁴ /m)	60.08	60.08	

CASE C03 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Modular ratio : $n_{enrobage}(G_1) =$	23.7	23.7	
Modular ratio : $n_{enrobage}(Q_c) =$	6.67	6.67	
Second moment of area (y-y axis) : $I_{el,y}(n = 6.67)$ (cm ⁴)	53 002	52 921	0.15%
Second moment of area (y-y axis) : $I_{el,y}(n = 23.7)$ (cm ⁴)	47 200	47 128	-0.15%
Second moment of area (y-y axis) : $I_{el,y}(n = 24.7)$ (cm ⁴)	47 083	47 005	-0.17%
Elastic neutral axis : $z_{el,y}(n = 6.67)$ (mm)	205	205	
Elastic neutral axis : $z_{el,y}(n = 23.7)$ (mm)	235	235	
Elastic neutral axis : $z_{el,y}(n = 24.7)$ (mm)	236	236	
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	777	775	0.26%

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	22	Rév. : D

CASE C03 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	60.5	60.5	
Bending moment : $M(G_1) =$ (kN.m)	189	189	
Shear force : $V(Q_c) =$ (kN)	13.3	13.3	
Bending moment : $M(Q_c) =$ (kN.m)	45.3	45.4	-0.22%

CASE C03 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.42	0.42	
Resistance to shear force : $\Gamma_V =$	0.10	0.10	
Resistance to MV interaction : $\Gamma_{MV} =$	0.42	0.42	
Critical moment to lateral torsional buckling : M_{cr} (kN.m)	903	914	-1.20%
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	0.66	0.63	4.76%

CASE C03 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	31.0	31.1	-0.32%
Deflection : $\delta(Q_c)$ (mm)	6.5	6.5	
Deflection : $\delta(SLS)$ (mm)	37.5	37.6	-0.27%

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	23	Rév. : D

5.5 CASE C04: Single span composite beam with partial encasement (composite phase)

5.5.1 Description

This test is based on the paper in the *Construction Métallique* review n°2023-4. It deals with a single span composite beam with a composite slab and partial encasement during its final phase in composite stage.

The values from the paper are used as references. Because the calculation method of the software is more accurate, some values of the review may not match perfectly.

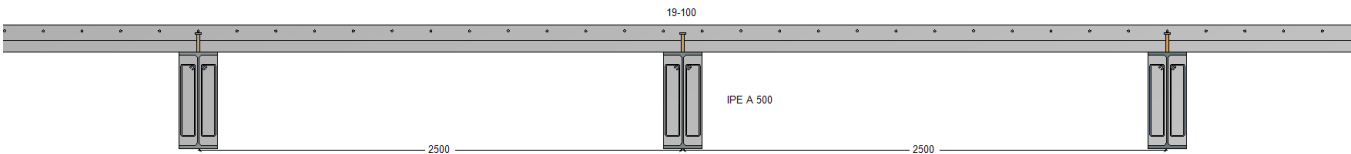


Figure 4 : Transversal view

5.5.2 Parameters

All parameters are the same as in Case 3, with the following specific parameters:

Definition of the connection:

- Studs: 19x100
- Two studs in each rib
- Unpropped beam during construction

Definition of loads:

- G1: self-weight only
- G2: surface load 1,0 kN/m²
- live load Q1: surface load 2,5 kN/m² ($\psi_2 = 0$)

Combination of loads:

- ULS: 1,35 (G1+G2) + 1,5 Q1
- SLS: (G1+G2) + Q1

Design parameters:

- Standard: Eurocodes first generation
- Design: According to the class of cross-section
- Relative humidity: 50 %
- Shrinkage deformation: 325×10^{-6}
- Age of concrete: $t_0 = 28 d$ for G2
 $t = 50 \times 365 d$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	24	Rév. : D

5.5.3 Verification of results

CASE C04 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	101	101.1	
Shear area : $A_v =$ (cm ²)	50.4	50.4	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	42 930	42 933	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 949	1 936	0.18%
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 728	1 728	
Elastic section modulus (z-z axis) : $W_{el,z} =$ (cm ³)	193,9	193.6	0.18%
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	1 946	1 946	
Plastic section modulus (z-z axis) : $W_{pl,z} =$ (cm ³)	301.6	298.3	1.15%
Torsional constant : $I_t =$ (cm ⁴)	64.18	64.28	-0.16%
Warping constant : $I_w =$ (cm ⁶)	1 125 000	1 125 230	
Mass by length unit : $m_{lin} =$ (kg/ml)	79,4	79,4	
Mass : $m =$ (kg)	994	992	0.16%
Total paint surface : $S =$ (cm ²)	217 643	217 643	
Paint surface (without upper face) : $S =$ (cm ²)	192 643	192 643	
Section factor : $M =$ (m ⁻¹)	172.0	172.2	-0.14%
Section factor (without upper face) : $M =$ (m ⁻¹)	152.2	152.4	-0.13%
Nominal yield strength : $f_y =$ (MPa)	355	355	
Axial resistance : $N_{t,Rd} =$ (kN)	3 595	3 589	0.15%
Shear resistance : $V_{pl,Rd} =$ (kN)	1 036	1 033	0.32%
Plastic resistance to bending : $M_{pl,Rd} =$ (kN.m)	692	691	0.16%
Elastic resistance to bending : $M_{el,Rd} =$ (kN.m)	614	613	0.24%
Section class for pure bending :	1	1	
Section class for pure compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	50.7	50.7	
Shear buckling resistance : $124\varepsilon =$	101	101	
Verification of web buckling required?	No	No	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	25	Rév. : D

CASE C04 RESULTS: PARTIAL ENCASEMENT	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	191	191	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	

CASE C04 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	164	164	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean tensile strength : $f_{ctm} =$ (MPa)	2.56	2.56	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 476	31 476	
Sec. moment of area per unit of width : $I_{pu} =$ (cm ⁴ /m)	60.08	60.08	
Shear resistance of studs : $P_{Rd} =$ (kN)	37.4	37.4	
Reduction factor : $k_t =$	0.50	0.50	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	26	Rév. : D

CASE C04 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Effective width of the slab : b_{eff} = (m)	2.50	2.50	
Modular ratio : $n_{slab}(G_1)$ =	—	—	
Modular ratio : $n_{encasement}(G_1)$ =	23.7	23.7	
Modular ratio : $n_{slab}(G_2)$ =	26.6	26.6	
Modular ratio : $n_{encasement}(G_2)$ =	23.7	23.7	
Modular ratio : $n_{slab}(Q_1)$ =	6.67	6.67	
Modular ratio : $n_{encasement}(Q_1)$ =	6.67	6.67	
Modular ratio : $n_{slab}(Q_c)$ =	∞	∞	
Modular ratio : $n_{encasement}(Q_c)$ =	6.67	6.67	
Modular ratio : $n_{slab}(SHC)$ =	25.2	25.2	
Modular ratio : $n_{slab}(SHC)$ =	24.7	24.7	
Moment of inertia about y-y axis : $I_{el,y}(G_1)$ (cm ⁴)	47 237	47 128	0.23%
Moment of inertia about y-y axis : $I_{el,y}(G_2)$ (cm ⁴)	99 895	99 778	0.12%
Moment of inertia about y-y axis : $I_{el,y}(Q_1)$ (cm ⁴)	142 875	142 812	0.14%
Moment of inertia about y-y axis : $I_{el,y}(Q_c)$ (cm ⁴)	53 094	52 921	0.33%
Moment of inertia about y-y axis : $I_{el,y}(SHC)$ (cm ⁴)	101 577	101 471	0.10 %
Elastic neutral axis : $z_{el,y}(G_1)$ (mm)	-235	-235	0.00%
Elastic neutral axis : $z_{el,y}(G_2)$ (mm)	-102.4	-102	0.39%
Elastic neutral axis : $z_{el,y}(Q_1)$ (mm)	9.38	9.51	-1.37%
Elastic neutral axis : $z_{el,y}(Q_c)$ (mm)	-205	-205	0.00%
Elastic neutral axis : $z_{el,y}(SHC)$ (mm)	-98.1	-97.9	0.20%
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	1 190	1 243	+4,4%

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	27	Rév. : D

CASE C04 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	60.5	60.5	
Bending moment : $M(G_1) =$ (kN.m)	189	189	
Shear force : $V(G_2) =$ (kN)	15.6	15.6	
Bending moment : $M(G_2) =$ (kN.m)	48.8	48.8	
Shear force : $V(Q_1) =$ (kN)	39.1	39.1	
Bending moment : $M(Q_1) =$ (kN.m)	122	122	
Shear force : $V(Q_c) =$ (kN)	13.3	13.3	
Bending moment : $M(Q_c) =$ (kN.m)	45.3	45.4	-0.22%
Shear force : $V(SHC) =$ (kN)	0	0	
Bending moment : $M(SHC) =$ (kN.m)	112.5	109.4	
Shear force : $V_{Ed} (ULS01) =$ (kN)	161	161	
Bending moment : $M_{Ed} (ULS01) =$ (kN.m)	504	504	
Shear force : $V_{Ed,const} (ULSC01) =$ (kN)	102	102	
Bending moment : $M_{Ed,const} (ULSC01) =$ (kN.m)	323	323	

CASE C04 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.42	0.426	
Resistance to shear force : $\Gamma_V =$	0.16	0.156	
Resistance to MV interaction : $\Gamma_{MV} =$	0.42	0.426	
Degree of shear connection : $\eta =$	0.78	0.773	
Minimum degree of connection : $\eta_{min} =$	0.63	0.625	
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	2.20	2.14	2.80%
Min. compression strut inclination angle : $\theta_{f,min} =$ (°)	27	27	
Resistance of the compression strut : $\Gamma_{sf} =$	0.61	0.59	3.39%
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	2.12	2.05	3.41%

CASE C04 RESULTS: ULS AT CONSTRUCTION STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0.42	0.42	
Resistance to shear force : $\Gamma_V =$	0.10	0.10	
Resistance to MV interaction : $\Gamma_{MV} =$	0.42	0.42	
Critical moment to lateral torsional buckling : M_{cr} (kN.m)	903	914	-1.20%
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	0.66	0.63	4.76%

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	28	Rév. : D

CASE C04 RESULTS: SLS ASSUMING FULL CONNECTION		Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$	(mm)	31.1	31.1	
Deflection : $\delta(G_2)$	(mm)	3.79	3.77	
Deflection : $\delta(Q_1)$	(mm)	6.62	6.63	
Deflection : $\delta(Q_c)$	(mm)	6.5	6.5	
Deflection : $\delta(SHC)$	(mm)	10.3	10.1	1.98%
Deflection : $\delta(SLS)$	(mm)	51.8	51.6	0.58%
Eigen frequency : $f(G + 0.2Q_1)$	(Hz)	4.6	4.7	-2.54%

5.6 CASE C05: Composite beam with cantilever span

5.6.1 Description

This case assesses a composite beam with a cantilever span at the left end.

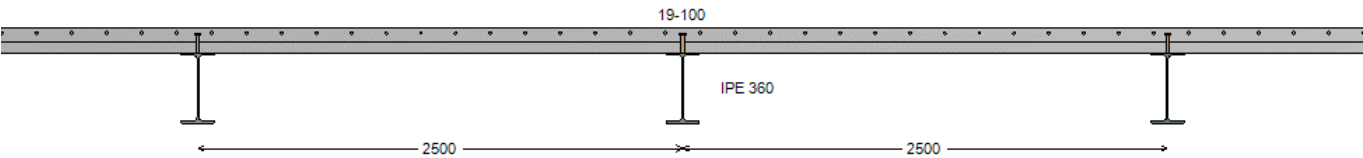


Figure 5 : Transversal view

5.6.2 Parameters

Definition of the beam:

Span lengths

Main span
 $L = 10,00$ m

☒ Left cantilever span
 $L_g = 3,00$ m

☐ Right cantilever span

☐ The slab is continuous at the right support

Spacings

☒ Intermediate beam
☐ Edge beam

Spacings
 $d_1 = 2,50$ m
 $d_2 = 2,50$ m

Slab openings

☐ Left opening
☐ Right opening

Definition of the slab:

- Composite slab
- Total depth: $t_d = 130$ mm
- Perpendicular ribs
- Steel profiled sheeting: Cofraplus 60, thickness 0,75 mm
- Prepunched sheeting

Definition of the profile:

- Hot rolled profile
- Profile: IPE 360
- Steel grade: S355 JR

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	30	Rév. : D

Definition of the concrete:

Normal concrete

Concrete class: C35/45

Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the connection:

Studs: 19x100

Two studs in each rib

Unpropped beam during construction

Definition of loads:

G1: self-weight only

G2: surface load 2 kN/m^2 on each span

live load Q1: surface load 16 kN/m^2 on each span

Combination of loads:

ULS: $1,35 (G1+G2) + 1,5 Q1$

SLS: $(G1+G2) + Q1$

Design parameters:

Standard: Eurocodes first generation

Design: According to the class of cross-section

Shear resistance: $\eta = 1,2$

Relative humidity: 50 %

Shrinkage deformation: 325×10^{-6}

Age of concrete: $t_0 = 28 d$ for G2

$t = 50 \times 365 d$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	31	Rév. : D

5.6.3 Verification of results

CASE C05 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	72,7	72,7	
Shear area : $A_v =$ (cm ²)	35,14	35,14	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	16 270	16 266	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 043	1 041	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	903,6	903,6	
Elastic section modulus (z-z axis) : $W_{el,z} =$ (cm ³)	122,8	122,5	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	1 019	1 019	
Plastic section modulus (z-z axis) : $W_{pl,z} =$ (cm ³)	191,1	188,9	
Torsional constant : $I_t =$ (cm ⁴)	37,32	37,44	
Warping constant : $I_w =$ (cm ⁶)	313 600	316 580	
Mass by length unit : $m_{lin} =$ (kg/ml)	57,1	57,1	
Nominal yield strenght : $f_y =$ (MPa)	355	355	
Tensile resistance : $N_{t,Rd} =$ (kN)	2 581,9	2582	
Shear resistance : $V_{pl,Rd} =$ (kN)	720,2	720	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	361,7	362	
Elastic moment resistant : $M_{el,Rd} =$ (kN.m)	320,8	321	
Section class for bending :	1	1	
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	41,83	41,8	
Shear buckling resistance : $72\varepsilon/\eta =$	48,82	48,8	
Verification of web buckling required?	No	No	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	32	Rév. : D

CASE C05 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	144	144	
Char. value of the compression strength : $f_{ck} =$ (MPa)	35	35	
Mean value of compression strength : $f_{cm} =$ (MPa)	43	43	
Mean tensile strength : $f_{ctm} =$ (MPa)		3,21	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	34 080	34 077	
Sec. moment of area per unit of width : $I_{pu} =$ (cm ⁴ /m)	44.37	44,37	
Shear resistance of the connector : $P_{Rd} =$ (kN)	41,13	41,13	
Reduction factor : $k_t =$	0,50	0,50	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	33	Rév. : D

CASE C05 RESULTS: ANALYSIS FOR LOAD CASES	Reference	ABCPMX	Dev.
Shear force at support A – central span : $V(G_1) =$ (kN)	35,88	35,88	
Shear force at support B – central span : $V(G_1) =$ (kN)	29,96	29,96	
Bending moment at support A : $M(G_1) =$ (kN.m)	-29,6	-29,6	
Max. bending moment in central span : $M(G_1) =$ (kN.m)	68,14	68,14	
Shear force at support A – central span : $V(G_2) =$ (kN)	27,25	27,25	
Shear force at support B – central span : $V(G_2) =$ (kN)	22,75	22,75	
Bending moment at support A : $M(G_2) =$ (kN.m)	-22,50	-22,50	
Max. bending moment in central span : $M(G_2) =$ (kN.m)	51,75	51,75	
Shear force at support A – central span : $V(Q_1\#1) =$ (kN)	218	218	
Shear force at support B – central span : $V(Q_1\#1) =$ (kN)	182	182	
Bending moment at support A : $M(Q_1\#1) =$ (kN.m)	-180	-180	
Max. bending moment in central span : $M(Q_1\#1) =$ (kN.m)	414	414	
Shear force at support A – central span : $V(Q_1\#2) =$ (kN)	200	200	
Shear force at support B – central span : $V(Q_1\#2) =$ (kN)	200	200	
Bending moment at support A : $M(Q_1\#2) =$ (kN.m)	0	0	
Max. bending moment in central span : $M(Q_1\#2) =$ (kN.m)	500	500	
Shear force at support A – central span : $V(Q_1\#3) =$ (kN)	18,0	18,0	
Shear force at support B – central span : $V(Q_1\#3) =$ (kN)	18,0	18,0	
Bending moment at support A : $M(Q_1\#3) =$ (kN.m)	-180,0	-180,0	
Max. bending moment in central span : $M(Q_1\#3) =$ (kN.m)	0	0	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	34	Rév. : D

CASE C05 RESULTS: ANALYSIS FOR ULS COMBINATION	Reference	ABCPMX	Dev.
Shear force at support A – central span : $V(ULS_1\#1) =$ (kN)	412,23	412,2	
Shear force at support B – central span : $V(ULS_1\#1) =$ (kN)	344,15	344,2	
Bending moment at support A : $M(ULS_1\#1) =$ (kN.m)	-340,37	-340,4	
Max. bending moment in central span : $M(ULS_1\#1) =$ (kN.m)	782,86	782,9	
Shear force at support A – central span : $V(ULS_1\#2) =$ (kN)	385,23	385,2	
Shear force at support B – central span : $V(ULS_1\#2) =$ (kN)	371,15	371,2	
Bending moment at support A : $M(ULS_1\#2) =$ (kN.m)	-70,37	-70,37	
Max. bending moment in central span : $M(ULS_1\#2) =$ (kN.m)	910,29	910,6	
Shear force at support A – central span : $V(ULS_1\#3) =$ (kN)	112,23	112,2	
Shear force at support B – central span : $V(ULS_1\#3) =$ (kN)	44,15	44,15	
Bending moment at support A : $M(ULS_1\#3) =$ (kN.m)	-340,37	-340,4	
Max. bending moment in central span : $M(ULS_1\#3) =$ (kN.m)	62,09	62,33	

CASE C05 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	1,575	1,575	
Resistance to shear force : $\Gamma_V =$	0,572	0,572	
Resistance to MV interaction : $\Gamma_{MV} =$	1,575	1,575	
Degree of shear connection ($M > 0$) : $\eta =$	0,431	0,428	
Minimum degree of connection ($M > 0$) : $\eta_{min} =$	0,505	0,505	
Degree of shear connection ($M < 0$) : $\eta =$	1	1,03	
Minimum degree of connection ($M < 0$) : $\eta_{min} =$	1	1	

CASE C05 RESULTS: TRANSVERSE REINFORCEMENTS IN THE MAIN SPAN	Reference	ABCPMX	Dev.
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	2,66	2,66	
Min. compression strut inclination angle : $\theta_{f,min} =$ (°)	36	36	
Resistance of the compression strut : $\Gamma_{sf} =$	0,464	0,46	
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	3,20	3,20	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	35	Rév. : D

5.7 CASE C05b: Composite beam with cantilever span

5.7.1 Description

This case assesses a composite beam with a cantilever span at the right end.

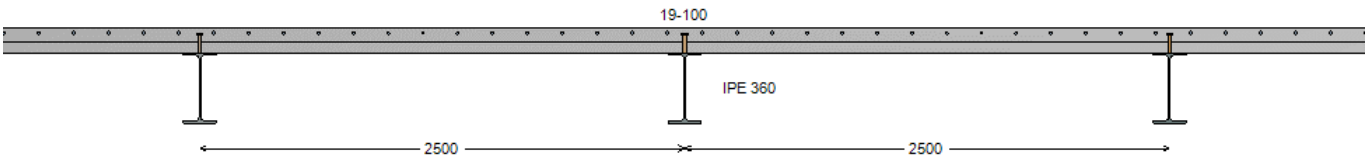


Figure 6 : Transversal view

5.7.2 Parameters

The case C05b is identical to Case C05, except that the cantilever is defined on the right end of the beam.

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	36	Rév. : D

5.7.3 Verification of results

CASE C05B RESULTS: ANALYSIS FOR LOAD CASES	Reference	ABCPMX	Dev.
Shear force at support A – central span : $V(G_1) =$ (kN)	29,96	29,96	
Shear force at support B – central span : $V(G_1) =$ (kN)	35,88	35,88	
Bending moment at support B : $M(G_1) =$ (kN.m)	-29,6	-29,6	
Max. bending moment in central span : $M(G_1) =$ (kN.m)	68,14	68,14	
Shear force at support A – central span : $V(G_2) =$ (kN)	22,75	22,75	
Shear force at support B – central span : $V(G_2) =$ (kN)	27,25	27,25	
Bending moment at support B : $M(G_2) =$ (kN.m)	-22,50	-22,50	
Max. bending moment in central span : $M(G_2) =$ (kN.m)	51,75	51,75	
Shear force at support A – central span : $V(Q_1\#1) =$ (kN)	182	182	
Shear force at support B – central span : $V(Q_1\#1) =$ (kN)	218	218	
Bending moment at support B : $M(Q_1\#1) =$ (kN.m)	-180	-180	
Max. bending moment in central span : $M(Q_1\#1) =$ (kN.m)	414	414	
Shear force at support A – central span : $V(Q_1\#2) =$ (kN)	200	200	
Shear force at support B – central span : $V(Q_1\#2) =$ (kN)	200	200	
Bending moment at support B : $M(Q_1\#2) =$ (kN.m)	0	0	
Max. bending moment in central span : $M(Q_1\#2) =$ (kN.m)	500	500	
Shear force at support A – central span : $V(Q_1\#3) =$ (kN)	18,0	18,0	
Shear force at support B – central span : $V(Q_1\#3) =$ (kN)	18,0	18,0	
Bending moment at support B : $M(Q_1\#3) =$ (kN.m)	-180,0	-180,0	
Max. bending moment in central span : $M(Q_1\#3) =$ (kN.m)	0	0	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	37	Rév. : D

CASE C05B RESULTS: ANALYSIS FOR ULS COMBINATION	Reference	ABCPMX	Dev.
Shear force at support A – central span : $V(ULS_1\#1) =$ (kN)	344,15	344,2	
Shear force at support B – central span : $V(ULS_1\#1) =$ (kN)	412,23	412,2	
Bending moment at support B : $M(ULS_1\#1) =$ (kN.m)	-340,37	-340,4	
Max. bending moment in central span : $M(ULS_1\#1) =$ (kN.m)	782,86	782,9	
Shear force at support A – central span : $V(ULS_1\#2) =$ (kN)	371,15	371,2	
Shear force at support B – central span : $V(ULS_1\#2) =$ (kN)	385,23	385,2	
Bending moment at support B : $M(ULS_1\#2) =$ (kN.m)	-70,37	-70,37	
Max. bending moment in central span : $M(ULS_1\#2) =$ (kN.m)	910,29	910,6	
Shear force at support A – central span : $V(ULS_1\#3) =$ (kN)	44,15	44,15	
Shear force at support B – central span : $V(ULS_1\#3) =$ (kN)	112,23	112,2	
Bending moment at support B : $M(ULS_1\#3) =$ (kN.m)	-340,37	-340,4	
Max. bending moment in central span : $M(ULS_1\#3) =$ (kN.m)	62,09	62,33	

CASE C05B RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	1,575	1,575	
Resistance to shear force : $\Gamma_V =$	0,572	0,572	
Resistance to MV interaction : $\Gamma_{MV} =$	1,575	1,575	
Degree of shear connection ($M > 0$) : $\eta =$	0,431	0,428	
Minimum degree of connection ($M > 0$) : $\eta_{min} =$	0,505	0,505	
Degree of shear connection ($M < 0$) : $\eta =$	1	1,03	
Minimum degree of connection ($M < 0$) : $\eta_{min} =$	1	1	

CASE C05B RESULTS: TRANSVERSE REINFORCEMENTS	Reference	ABCPMX	Dev.
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	2,66	2,66	
Min. compression strut inclination angle : $\theta_{f,min} =$ (°)	36	36	
Resistance of the compression strut : $\Gamma_{sf} =$	0,464	0,46	
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	3,20	3,20	

5.8 CASE C06: Composite beam with a solid slab

5.8.1 Description

This case assesses a composite beam with a solid slab.

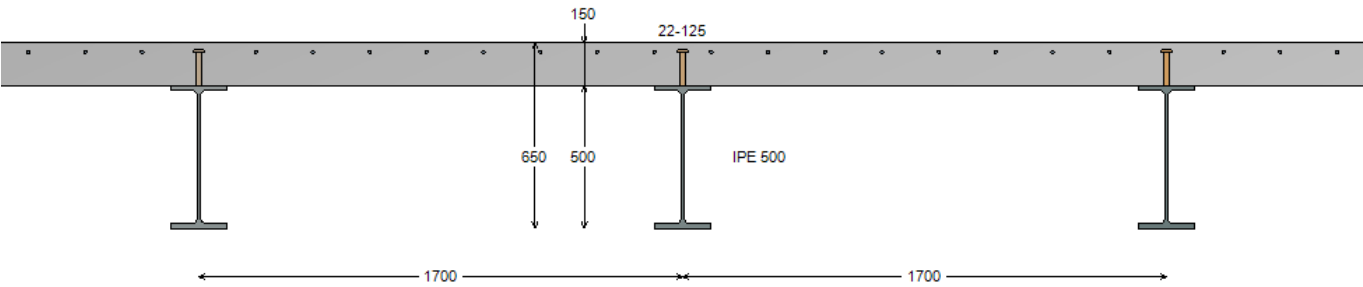


Figure 7 : Transversal view

5.8.2 Parameters

Definition of the beam:

Definition of the beam

Span lengths

Main span
L= 14,00 m

☐ Left cantilever span
☐ The slab is continuous at the left support

☐ Right cantilever span
☐ The slab is continuous at the right support

Slab openings

☐ Left opening
☐ Right opening

Spacings

☒ Intermediate beam
☐ Edge beam

Spacings
d₁ = 1,70 m
d₂ = 1,70 m

14,00

1,70

1,70

Cancel

OK

Definition of the slab:

Solid slab

Total depth: $t_d = 150 \text{ mm}$

Definition of the profile:

Hot rolled profile

Profile: IPE 500

Steel grade: S460 EC3

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	39	Rév. : D

Definition of the concrete:

Normal concrete

Concrete class: C25/30

Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the connection:

Studs: 22x125

Two studs every 350 mm along the beam

The beam is fully propped during construction

Definition of loads:

G1: self-weight only

G2: surface load 2 kN/m^2

live load Q1: surface load 8 kN/m^2 ($\psi_2 = 0,4$)

Combination of loads:

ULS: $1,35 (G1+G2) + 1,5 Q1$

SLS: $(G1+G2) + Q1$

Design parameters:

Standard: Eurocodes first generation

Design: According to the class of cross-section

Relative humidity: 50 %

Shrinkage deformation: 325×10^{-6}

Age of concrete: $t_0 = 28 d$ for G2

$t = 50 \times 365 d$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	40	Rév. : D

5.8.3 Verification of results

CASE C06 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	115,5	115,5	
Shear area : $A_v =$ (cm ²)	59,87	59,87	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	48 200	48 199	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 930	1 928	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	2194	2 194	
Total mass : $m =$ (kg)	1270	1270	
Nominal yield strenght : $f_y =$ (MPa)	460	460	
Tensile resistance : $N_{t,Rd} =$ (kN)	5314	5314	
Shear resistance : $V_{pl,Rd} =$ (kN)	1590	1590	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	1009	1009	
Section class for bending :	1	1	
Section class for compression :	4	4	

CASE C06 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	159	159	
Char. value of the compression strength : $f_{ck} =$ (MPa)	25	25	
Mean value of compression strength : $f_{cm} =$ (MPa)	33	33	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	31 480	31 476	
Shear resistance of the connector : $P_{Rd} =$ (kN)	99,6	99,6	

CASE C06 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Effective width of the slab : $b_{eff} =$ (m)	1,7	1,7	
Modular ratio : $n(G_{1,2}) =$	26,7	26,7	
Modular ratio : $n(Q_1) =$	6,67	6,67	
Modular ratio : $n(SH) =$	25,3	25,3	
Second moment of area (y-y axis) : $I_{el,y}(n = 6.67)$ (cm ⁴)	148 643	149 064	
Second moment of area (y-y axis) : $I_{el,y}(n = 26.9)$ (cm ⁴)	105 214	105 256	
Second moment of area (y-y axis) : $I_{el,y}(n = 25.5)$ (cm ⁴)	106 945	106 974	
Plastic neutral axis : $z_{pl,y}$ (mm)	-9,4	-9,3	
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	1501	1501	
Reduction factor: β	0,943	0,943	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	41	Rév. : D

CASE C06 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(ULS) =$ (kN)	242,4	242,4	
Bending moment : $M(ULS) =$ (kN.m)	848,5	848,5	

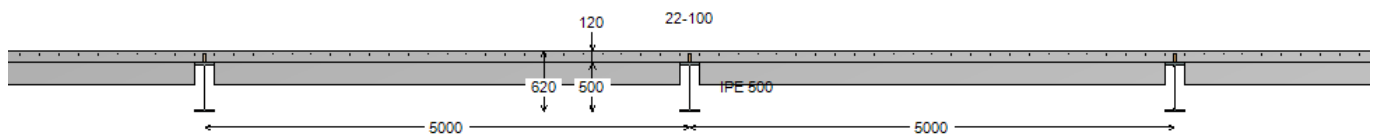
CASE C06 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_M =$	0,565	0,565	
Resistance to shear force : $\Gamma_V =$	0,152	0,152	
Degree of shear connection : $\eta =$	1	1,1	
Potential area of shear failure a-a			
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	1,84	1,84	
Min. compression strut inclination angle : $\theta_{f,min} =$ (°)	27	27	
Resistance of the compression strut : $\Gamma_{sf} =$	0,504	0,50	
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	3,23	3,23	
Potential area of shear failure b-b			
Longitudinal shear force : $\tau_{Ed} =$ (MPa)	1,74	1,74	
Min. compression strut inclination angle : $\theta_{f,min} =$ (°)	27	27	
Resistance of the compression strut : $\Gamma_{sf} =$	0,478	0,48	
Area of reinf. by length unit : $A_{sf}/s_f =$ (cm ² /m)	3,23	6,67	

CASE C06 RESULTS: SLS ASSUMING FULL SHEAR CONNECTION	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	16,21	16,2	
Deflection : $\delta(G_2)$ (mm)	7,71	7,69	
Deflection : $\delta(Q_1)$ (mm)	25,4	25,3	

5.9 CASE C07: Composite beam with a composite slab and Cofraplus 220

5.9.1 Description

This case assesses a composite beam with a composite slab using a Cofraplus 220.



5.9.2 Parameters

Definition of the beam:

Definition of the beam

Span lengths

Main span
L= 14.00 m

☐ Left cantilever span

☐ The slab is continuous at the left support

☐ Right cantilever span

☐ The slab is continuous at the right support

Slab openings

☐ Left opening

☐ Right opening

Spacings

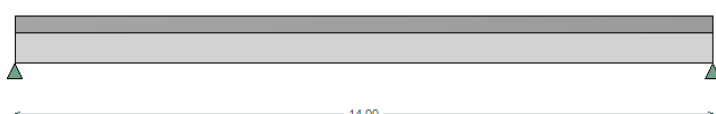
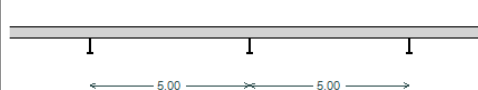
☒ Intermediate beam

☐ Edge beam

Spacings

d₁ = 5.00 m

d₂ = 5.00 m

Cancel OK

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	43	Rév. :

Definition of the slab:

Composite slab with a Cofraplus 220 profiled steel sheeting

Definition of the slab

General

Type

Composite slab

Thickness

☒ $t_d = 340$ mm

☐ $t_c = 120$ mm

Concrete

☐ Light concrete

Class

C20/25

$\rho_c = 2400$ kg/m³

Longitudinal reinforcement

First layer of reinforcement

Diameter

$\phi_{s1} = 12.00$ mm

Spacing

$e_{s1} = 200.0$ mm

Location

$z_{s1} = 32.00$ mm

Reinforcement steel

Class

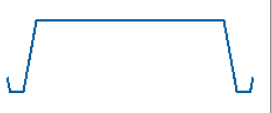
B500

$f_{sk} = 500$ MPa

Profiled steel sheeting

Name

Cofraplus_220 1.25



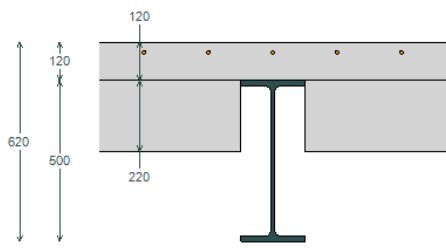
Height

$h_p = 220.0$ mm

Modify the steel sheeting

Rib orientation

☒ Perpendicular



Cancel

OK

Definition of the profile:

Hot rolled profile

Profile: IPE 500
Steel grade: S420 EC3

Definition of the concrete:

Normal concrete

Concrete class: C20/25
Density: $\rho_c = 2400$ kg/m³

Definition of the connection:

Studs: 22x100
Two studs every 250 mm along the beam
The beam is fully propped during construction

Definition of loads:

G1: self-weight only
G2: surface load 1 kN/m²
live load Q1: surface load 3 kN/m² ($\psi_2 = 0$)

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	44	Rév. : D

Combination of loads:

ULS: 1,35 (G1+G2) + 1,5 Q1
SLS: (G1+G2) + Q1

Design parameters:

Standard: Eurocodes first generation
Design: According to the class of cross-section
Relative humidity: 50 %
Shrinkage deformation: 325×10^{-6}
Age of concrete: $t_0 = 28 d$ for G2
 $t = 50 \times 365 d$

5.9.3 Verification of results

CASE C07 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	115,5	115,5	
Shear area : $A_v =$ (cm ²)	59,87	59,87	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	48 200	48 199	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 930	1 928	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	2194	2 194	
Total mass : $m =$ (kg)	1270	1270	
Nominal yield strenght : $f_y =$ (MPa)	420	420	
Tensile resistance : $N_{t,Rd} =$ (kN)	4 852	4 852	
Shear resistance : $V_{pl,Rd} =$ (kN)	1 452	1 452	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	921	922	
Section class for bending :	1	1	
Section class for compression :	4	4	

CASE C07 RESULTS: SLAB	Reference	ABCPMX	Dev.
Notional size (EN 1992-1-1 Annex B) : $h_0 =$ (mm)	240	240	
Char. value of the compression strength : $f_{ck} =$ (MPa)	20	20	
Mean value of compression strength : $f_{cm} =$ (MPa)	28	28	
Mean secant modulus of elasticity : $E_{cm} =$ (MPa)	29 960	29 962	
Shear resistance of the connector : $P_{Rd} =$ (kN)	86,9	86,9	
Mass of the slab: (kg/m)	1 880	1 880	
Weight of the slab: (kN/m ²)	19,26	19,26	

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	45	Rév. : D

CASE C07 RESULTS: SECTION PROPERTIES AT MID-SPAN	Reference	ABCPMX	Dev.
Effective width of the slab : b_{eff} = (m)	3,5		
Modular ratio : $n(G_{1,2})$ =	28,4	28,4	
Modular ratio : $n(Q_1)$ =	7,01	7,01	
Modular ratio : $n(SH)$ =	26,9	26,9	
Second moment of area (y-y axis) : $I_{el,y}(n = 7,01)$ (cm ⁴)	148 444	148 446	
Second moment of area (y-y axis) : $I_{el,y}(n = 26,9)$ (cm ⁴)	113 880	113 889	
Second moment of area (y-y axis) : $I_{el,y}(n = 28,4)$ (cm ⁴)	112 303	112 330	
Plastic neutral axis : $z_{pl,y}$ (mm)	0,6	0,6	
Plastic resistance to bending moment : $M_{pl,Rd}$ (kN.m)	1459	1 459	
Reduction factor: β	0,973	0,973	

CASE C07 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(ULS)$ = (kN)	395	395	
Bending moment : $M(ULS)$ = (kN.m)	1383	1383	

CASE C07 RESULTS: ULS AT FINAL STAGE	Reference	ABCPMX	Dev.
Resistance to bending moment: Γ_M =	0,948	0,948	
Resistance to shear force: Γ_V =	0,272	0,272	
Degree of shear connection: η =	1,02	1,023	
Potential area of shear failure a-a			
Longitudinal shear force : τ_{Ed} = (MPa)	2,852	2,85	
Min. compression strut inclination angle : $\theta_{f,min}$ = (°)	27	27	
Resistance of the compression strut : Γ_{sf} =	0,958	0,96	
Area of reinf. by length unit : A_{sf}/s_f = (cm ² /m)	4,01	4,01	
Potential area of shear failure b-b			
Longitudinal shear force : τ_{Ed} = (MPa)	2,510	2,51	
Min. compression strut inclination angle : $\theta_{f,min}$ = (°)	27	27	
Resistance of the compression strut : Γ_{sf} =	0,843	0,84	
Area of reinf. by length unit : A_{sf}/s_f = (cm ² /m)	8,15	8,15	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	46	Rév. : D

CASE C07 RESULTS: SLS ASSUMING FULL SHEAR CONNECTION		Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$	(mm)	42,7	42,8	
Deflection : $\delta(G_2)$	(mm)	10,6	10,6	
Deflection : $\delta(Q_1)$	(mm)	24,1	24,1	

6 TESTS CASES – STEEL BEAMS

6.1 CASE S01: Steel beam with 2 cantilever spans

6.1.1 Description

This test deals with a steel beam with both left and right cantilevers.

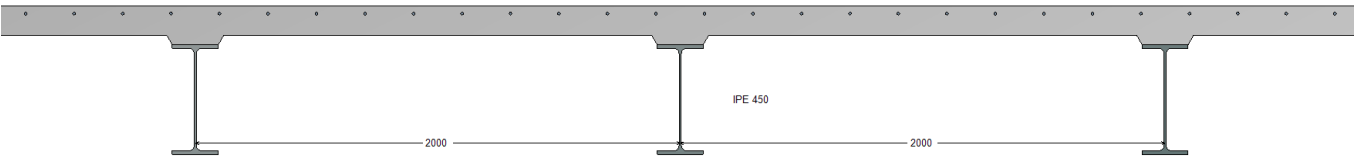


Figure 8 : Transversal view

6.1.2 Parameters

Definition of the beam:

Span lengths

Main span

L = 10,00 m

☒ Left cantilever span

L_g = 3,00 m

☒ Right cantilever span

L_d = 3,00 m

Spacings

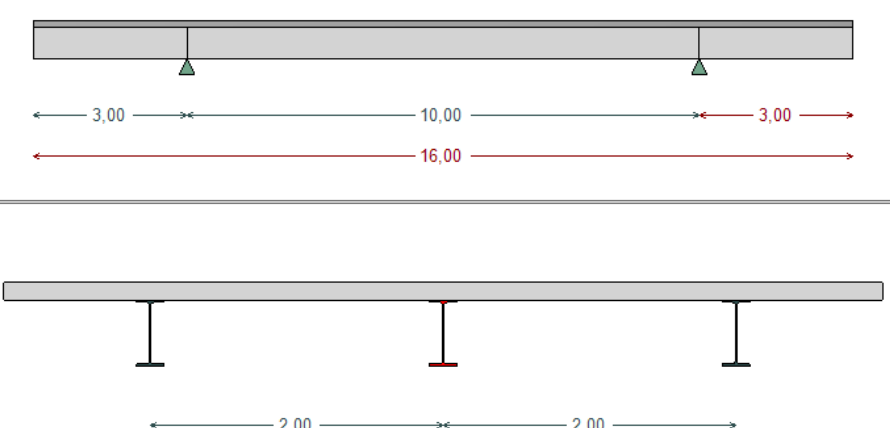
☒ Intermediate beam

☐ Edge beam

Spacings

d₁ = 2,00 m

d₂ = 2,00 m



The diagram illustrates the beam configuration with two rows. The top row shows the span lengths: a main span of 10,00 m between two supports, with left and right cantilever spans of 3,00 m each. The total length is 16,00 m. The bottom row shows the spacings between the three supports, both d₁ and d₂ are 2,00 m.

Definition of the profile:

Hot rolled profile

Profile:

IPE 450

Steel grade:

S355 M/ML

Definition of the slab:

General

TypeSolid slab

Thickness $t_d = 120$ mm

Haunch $t_h = 40$ mm

Concrete

☐ Light concrete

ClassC20/25

$\rho_c = 2500$ kg/m3

Longitudinal reinforcement



First layer of reinforcement

Diameter $\phi_{s1} = 12.00$ mm

Spacing $e_{s1} = 200.0$ mm

Location $z_{s1} = 32.00$ mm

Reinforcement steel

ClassB500

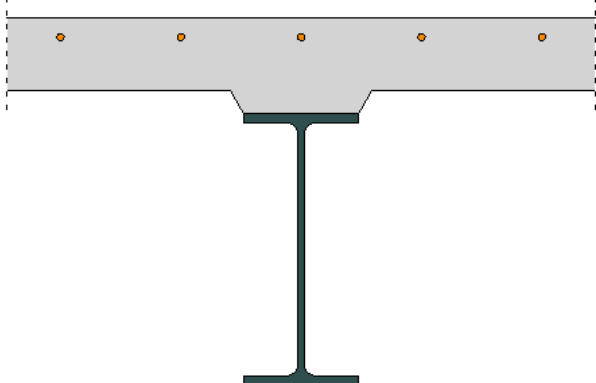
$f_{sk} = 500$ MPa

610

120

40

450



Lateral restraint of the profile:

One lateral restraints at mid-span of the central span, for both flanges

Definition of loads:

G1:	self-weight only	
live load Q1:	left cantilever span:	distributed load from 0 (left end) to 1 kN/m (right end)
	central span:	distributed load from 1 (left end) to 3 kN/m (right end)
	right cantilever span:	distributed load from 3 (left end) to 5 kN/m (right end)

Other parameters:

$\eta = 1,2$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	49	Rév. : D

6.1.3 Verification of results

CASE S01 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	98.8	98.8	
Shear area : $A_v =$ (cm ²)	50.8	50.8	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	33 740	33 743	0.01%
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 676	1 672	0.24%
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	1 500	1 500	
Elastic section modulus (z-z axis) : $W_{el,z} =$ (cm ³)	176	176	
Plastic section modulus (y-y axis) : $W_{pl,y} =$ (cm ³)	1 702	1 702	
Plastic section modulus (z-z axis) : $W_{pl,z} =$ (cm ³)	276	272.8	1.17%
Torsional constant : $I_t =$ (cm ⁴)	66.66	66.75	0.13%
Warping constant : $I_w =$ (cm ⁶)	780 739	791 005	1.30%
Mass by length unit : $m_{lin} =$ (kg/ml)	77.7	77.6	0.13%
Mass : $m =$ (kg)	1 243	1 241	0.16%
Total paint surface : $S =$ (m ²)	25,68	25,68	
Paint surface (without upper face) : $S =$ (m ²)	22,64	22,64	
Section factor : $M =$ (m ⁻¹)	162.4	162.4	
Section factor (without upper face) : $M =$ (m ⁻¹)	143.2	143.2	
Nominal yield strength : $f_y =$ (MPa)	355	355	
Axial resistance : $N_{t,Rd} =$ (kN)	3 508	3 508	
Shear resistance : $V_{pl,Rd} =$ (kN)	1 402	1 402	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	604	604	
Elastic moment resistant : $M_{el,Rd} =$ (kN.m)	532	532	
Section class for bending :	1	1	
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	44,8	44,8	
Shear buckling resistance : $72\varepsilon/\eta =$	48.8	48.8	
Verification of web buckling required?	No	No	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	50	Rév. : D

CASE S01 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	34.4	34.2	0.58%
Bending moment : $M(G_1) =$ (kN.m)	55.1/-31	54.8/-30.8	0.73%/0.65%
Shear force : $V(Q_1\#1) =$ (kN)	13.5	13.5	
Bending moment : $M(Q_1\#1) =$ (kN.m)	14.7/-19.5	14.7/-19.5	
Shear force : $V(Q_1\#2) =$ (kN)	11.7	11.7	
Bending moment : $M(Q_1\#2) =$ (kN.m)	25.2	25.2	
Shear force : $V(Q_1\#3) =$ (kN)	12	12	
Bending moment : $M(Q_1\#3) =$ (kN.m)	-19.5	-19.5	
Shear force : $V_{Ed,ULS\#1} =$ (kN)	66.7	66.4	0.45%
Bending moment : $M_{Ed,ULS\#1} =$ (kN.m)	96.2/-71.1	95.8/-70.8	0.52%/0.42%
Shear force : $V_{Ed,ULS\#2} =$ (kN)	64.0	63.7	0.47%
Bending moment : $M_{Ed,ULS\#2} =$ (kN.m)	112/-41.8	111/-41.6	0.90%/0.48%
Shear force : $V_{Ed,ULS\#3} =$ (kN)	49.2	48.9	0.61%
Bending moment : $M_{Ed,ULS\#3} =$ (kN.m)	59.0/-71.1	58.6/-70.8	0.85%/0.42%

CASE S01 RESULTS: ULS IN THE CENTRAL SPAN	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_{M,ULS\#1} =$	0.16	0.159	
Resistance to shear force : $\Gamma_{V,ULS\#1} =$	0.06	0.064	
Resistance to MV interaction : $\Gamma_{MV,ULS\#1} =$	0.16	0.159	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS\#1}$ (kN.m)	521.97	523.4	-0.27%
Resistance to lateral torsional buckling : $\Gamma_{LT,ULS\#1} =$	0.29	0.288	
Resistance to bending moment : $\Gamma_{M,ULS\#2} =$	0.19	0.185	
Resistance to shear force : $\Gamma_{V,ULS\#2} =$	0.06	0.061	
Resistance to MV interaction : $\Gamma_{MV,ULS\#2} =$	0.19	0.185	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS\#2}$ (kN.m)	546.2	545.3	0.53%
Resistance to lateral torsional buckling : $\Gamma_{LT,ULS\#2} =$	0.33	0.327	
Resistance to bending moment : $\Gamma_{M,ULS\#3} =$	0.12	0.117	
Resistance to shear force : $\Gamma_{V,ULS\#3} =$	0.05	0.047	
Resistance to MV interaction : $\Gamma_{MV,ULS\#3} =$	0.12	0.117	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS\#3}$ (kN.m)	496.6	497.1	-0.10%
Resistance to lateral torsional buckling : $\Gamma_{LT,ULS\#3} =$	0.22	0.220	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	51	Rév. : D

CASE S01 RESULTS: SLS		Reference	ABCPMX	Dev.
Deflection central span : $\delta(G_1)$	(mm)	7.19	7.14	
Deflection central span : $\delta(Q_1\#1)$	(mm)	1.83	1.83	0.00%
Deflection central span : $\delta(Q_1\#2)$	(mm)	3.68	3.68	0.00%
Deflection right span : $\delta(Q_1\#3)$	(mm)	3.50	3.50	
Deflection central span : $\delta(SLS\#1)$	(mm)	9.01	8.96	
Deflection central span : $\delta(SLS\#2)$	(mm)	10.9	10.8	
Deflection central span : $\delta(SLS\#3)$	(mm)	5.34	5.30	

6.2 CASE S02: Steel beam with welded profile

6.2.1 Description

This test deals with a steel beam with welded profile (double symmetric).

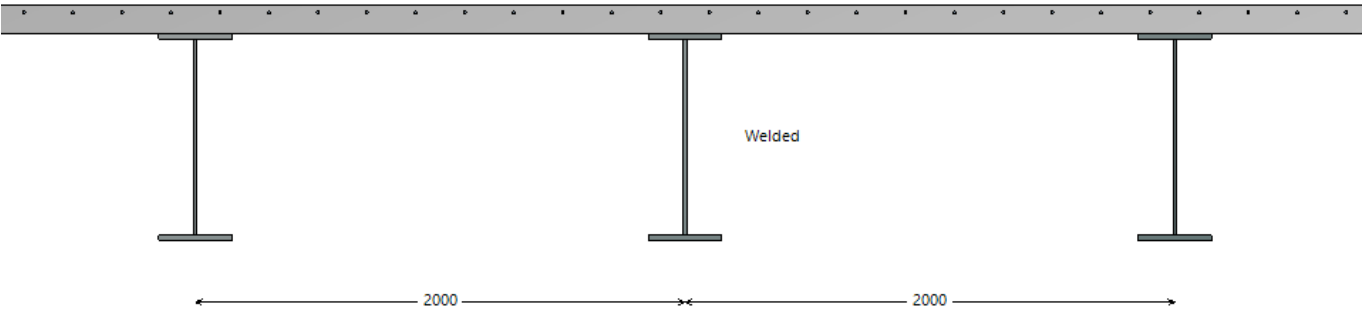


Figure 9 : Transversal view

6.2.2 Parameters

Definition of the beam:

Single span
Span length: 15 m

Definition of the profile:

Welded profile
Upper flange: 300x20
Web: 800x12
Lower flange: 300x20
Steel grade: S355 JR

Definition of loads:

G1: distributed load: 12 kN/m
live load Q1: distributed load: 20 kN/m
live load Q2: concentrated load: 15 kN at mid span

Lateral restraints:

2 lateral restraints of both flange at 1/3 and 2/3 of the span length

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	53	Rév. : D

Partial factors and national factors:

$$\gamma_G = 1,4$$

$$\gamma_Q = 1,6$$

$$\gamma_{M0} = 1,05$$

$$\gamma_{M1} = 1,1$$

$$\gamma_{M2} = 1,25$$

$$\psi_0 = 0,7$$

$$\eta = 1,2$$

Combination of loads:

$$\text{ULS-01:} \quad \gamma_G G + \gamma_Q Q_1 + \gamma_Q \psi_0 Q_2$$

$$\text{ULS-02:} \quad \gamma_G G + \gamma_Q \psi_0 Q_1 + \gamma_Q Q_2$$

$$\text{SLS:} \quad G + Q_1 + Q_2$$

Design parameters:

Standard: Eurocodes first generation

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	54	Rév. : D

6.2.3 Verification of results

CASE S02 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	216	216	
Shear area : $A_v =$ (cm ²)	115,2	115,2	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	252 960	252 960	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	9012	9012	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	6023	6023	
Mass by length unit : $m_{lin} =$ (kg/ml)	170	170	
Mass : $m =$ (kg)	2543	2543	
Total paint surface : $S =$ (cm ²)	42,84	42,84	
Paint surface (without upper face) : $S =$ (cm ²)	38,34	38,34	
Nominal yield strength flanges : $f_y =$ (MPa)	345	345	
Nominal yield strength web : $f_y =$ (MPa)	355	355	
Axial resistance : $N_{t,Rd} =$ (kN)	7189	7189	
Shear resistance : $V_{pl,Rd} =$ (kN)	2249	2249	
Plastic bending resistance (y-y axis) : $M_{pl,Rd} =$ (kNm)	2266	2266	
Elastic bending resistance (y-y axis) : $M_{el,Rd} =$ (kNm)	1979	1979	
Section class for bending :	2	2	
Section class for compression :		4	
Shear buckling resistance : $h_w/t_w =$	66,67	66,67	
Shear buckling resistance : $72\varepsilon/\eta =$	48,82	48,82	
Verification of web buckling required?	Yes	Yes	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	55	Rév. : D

CASE S02 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	90	90	
Bending moment : $M(G_1) =$ (kN.m)	337,5	337,5	
Shear force : $V(Q_1) =$ (kN)	150	150	
Bending moment : $M(Q_1) =$ (kN.m)	562.5	562.5	
Shear force : $V(Q_2) =$ (kN)	7,5	7,5	
Bending moment : $M(Q_2) =$ (kN.m)	56,3	56,3	
Shear force : $V_{Ed,ULS-1} =$ (kN)	374,4	374,4	
Bending moment : $M_{Ed,ULS-1} =$ (kN.m)	1435,5	1435,5	
Shear force : $V_{Ed,ULS-2} =$ (kN)	306	306	
Bending moment : $M_{Ed,ULS-2} =$ (kN.m)	1192,5	1192,5	

CASE S02 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,634	0,634	
Resistance to shear force : Γ_V	0,166	0,166	
Resistance to shear buckling : Γ_{Vb}	0,239	0,239	
Resistance to MV interaction : Γ_{MV}	-	-	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS-1}$ (kN.m)	4163	4163	
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	1,09	1,09	
Minimum design size of welds: a_w (mm)	0,75	0,75	

CASE S02 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G)$ (mm)	14,9	14,9	
Deflection : $\delta(Q_1)$ (mm)	24,8	24,8	
Deflection : $\delta(Q_2)$ (mm)	1,99	1,98	
Deflection : $\delta(SLS)$ (mm)	41,7	41,7	
Eigen frequency : $f(G + 0,2 Q_1)$ (Hz)	3,98	3,98	
Modal mass: (kg)	12 232	12 232	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	56	Rév. : D

6.3 CASE S02b: Steel beam with welded profile

6.3.1 Description

This test deals with a steel beam with welded profile (double symmetric). An elastic design is triggered by the class 3 section

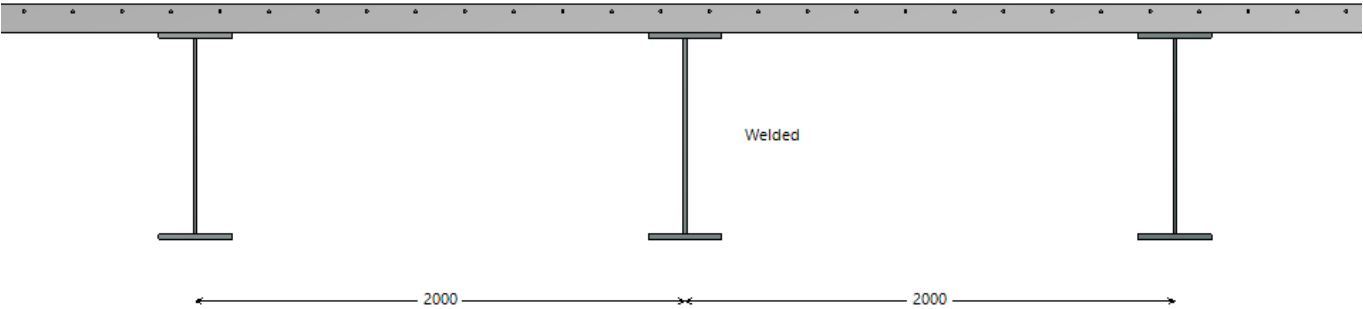


Figure 10 : Transversal view

6.3.2 Parameters

All parameters are the same as in Case S02, with the following modifications:

Definition of the profile:

Welded profile	
Upper flange:	300x20
Web:	800x12
Lower flange:	300x20
Steel grade:	S460 JR

Definition of loads:

G1:	distributed load:	12 kN/m
live load Q1:	distributed load:	20 kN/m
live load Q2:	concentrated load:	30 kN at mid span

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	57	Rév. : D

6.3.3 Verification of results

CASE S02B RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	216	216	
Shear area : $A_v =$ (cm ²)	115,2	115,2	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	252 960	252 960	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	9012	9012	
Elastic section modulus (y-y axis) : $W_{el,y} =$ (cm ³)	6023	6023	
Mass by length unit : $m_{lin} =$ (kg/ml)	170	170	
Mass : $m =$ (kg)	2543	2543	
Total paint surface : $S =$ (cm ²)	42,84	42,84	
Paint surface (without upper face) : $S =$ (cm ²)	38,34	38,34	
Nominal yield strength flanges : $f_y =$ (MPa)	440	440	
Nominal yield strength web : $f_y =$ (MPa)	460	460	
Axial resistance : $N_{t,Rd} =$ (kN)	9234	9234	
Shear resistance : $V_{pl,Rd} =$ (kN)	2914	2914	
Plastic bending resistance (y-y axis) : $M_{el,Rd} =$ (kNm)	2524	2524	
Elastic bending resistance (y-y axis) : $M_{pl,Rd} =$ (kNm)	2903	2903	
Section class for bending :	3	3	
Shear buckling resistance : $h_w/t_w =$	66,67	66,67	
Shear buckling resistance : $72\varepsilon/\eta =$	42,89	42,89	
Verification of web buckling required?	Yes	Yes	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	58	Rév. : D

CASE S02B RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	90	90	
Bending moment : $M(G_1) =$ (kN.m)	337,5	337,5	
Shear force : $V(Q_1) =$ (kN)	150	150	
Bending moment : $M(Q_1) =$ (kN.m)	562,5	562,5	
Shear force : $V(Q_2) =$ (kN)	15	15	
Bending moment : $M(Q_2) =$ (kN.m)	112,5	112,5	
Shear force : $V_{Ed,ULS-1} =$ (kN)	382,8	382,8	
Bending moment : $M_{Ed,ULS-1} =$ (kN.m)	1498,5	1498,5	
Shear force : $V_{Ed,ULS-2} =$ (kN)	318	318	
Bending moment : $M_{Ed,ULS-2} =$ (kN.m)	1282,5	1282,5	

CASE S01 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,594	0,594	
Resistance to shear force : Γ_V	0,131	0,131	
Resistance to shear buckling : Γ_{Vb}	0,215	0,215	
Resistance to MV interaction : Γ_{MV}	-	-	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS-1}$ (kN.m)	4189	4189	
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	1,07	1,07	
Minimum design size of welds: a_w (mm)	0,75	0,73	

CASE S02 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G)$ (mm)	14,9	14,9	
Deflection : $\delta(Q_1)$ (mm)	24,8	24,8	
Deflection : $\delta(Q_2)$ (mm)	3,97	3,97	
Deflection : $\delta(SLS)$ (mm)	43,7	43,7	
Eigen frequency : $f(G + 0,2 Q_1)$ (Hz)	3,98	3,98	
Modal mass: (kg)	12 232	12 232	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	59	Rév. : D

6.4 CASE S03: Steel beam with welded profile and MV interaction

6.4.1 Description

This test deals with a steel beam with welded profile and a MV interaction.

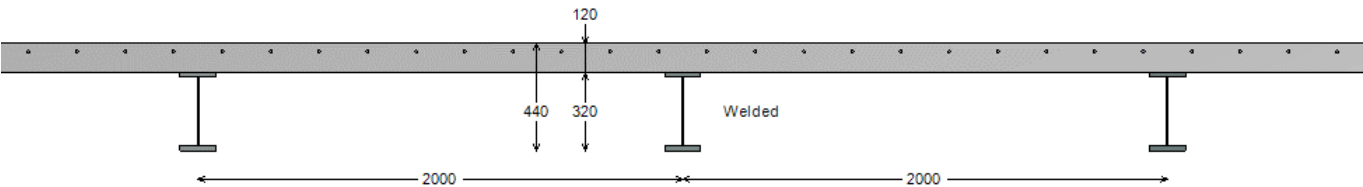


Figure 11 : Transversal view

6.4.2 Parameters

Definition of the beam:

- Single span with a cantilever span on the left
- Span length: 9 m
- Cantilever length: 2,5 m

Definition of the profile:

- Welded profile
- Upper flange: 150x18
- Web: 280x6
- Lower flange: 150x22
- Steel grade: S275 JR

Definition of the slab:

- solid slab: $t_d = 120\text{ mm}$

Definition of loads:

- live load Q1: distributed load on the cantilever span: 75 kN/m

Lateral restraints:

- 2 lateral restraints of both flange at 1/3 and 2/3 of the span length

Partial factors and national factors:

- $\gamma_{M0} = 1,0$
- $\gamma_{M1} = 1,0$
- $\gamma_{M2} = 1,25$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	60	Rév. : D

$$\eta = 1,0$$

Combination of loads:

ULS-01: Q_1

SLS: Q_1

Design parameters:

Standard: Eurocodes first generation

Working draft

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	61	Rév. : D

6.4.3 Verification of results

CASE S03 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	76,80	76,80	
Shear area : $A_v =$ (cm ²)	16,8	16,8	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	14 517	14 517	
Location of the elastic neutral axis: $z_{y,ENA} =$ (mm)	170,5	170,5	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 125	1 126	
Torsion constant : $I_t =$ (cm ⁴)	78,91	78,91	
Warping factor : $I_w =$ (cm ⁶)	250593	250594	
Mass by length unit : $m_{lin} =$ (kg/ml)	60	60	
Mass : $m =$ (kg)	693	693	
Total paint surface : $S =$ (cm ²)	14,12	14,12	
Paint surface (without upper face) : $S =$ (cm ²)	12,40	12,40	
Nominal yield strength flanges : $f_y =$ (MPa)	275	275	
Nominal yield strength web : $f_y =$ (MPa)	275	275	
Axial resistance : $N_{t,Rd} =$ (kN)	2 112	2 112	
Shear resistance : $V_{pl,Rd} =$ (kN)	267,0	267	
Plastic bending resistance (y-y axis) : $M_{pl,Rd} =$ (kNm)	276	276	
Location of the plastic neutral axis: $z_{y,PNA} =$ (mm)	208	208	
Elastic bending resistance (y-y axis) : $M_{el,Rd} =$ (kNm)	234	234	
Section class for bending :	1	1	
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	46,67	46,7	
Shear buckling resistance : $72\varepsilon/\eta =$	66,56	66,56	
Verification of web buckling required?	No	No	

CASE S03 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(Q_1) =$ (kN)	187,5	187,5	
Bending moment : $M(Q_1) =$ (kN.m)	-234,4	-234,4	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	62	Rév. : D

CASE S03 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,850	0,850	
Resistance to shear force : Γ_V	0,703	0,703	
Resistance to shear buckling : Γ_{Vb}	-	-	
Resistance to MV interaction : Γ_{MV}	0,869	0,869	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS-1}$ (kN.m)	632,5	642,8	+1,6%
Resistance to lateral torsional buckling : Γ_{LT} =	1,27	1,26	
Minimum design size of welds upper flange: a_w (mm)	1,40	1,40	
Minimum design size of welds lower flange: a_w (mm)	1,47	1,47	

CASE S03 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection at the end of the cantilever : $\delta(Q_1)$ (mm)	69,7	69,7	
Deflection at mid span : $\delta(Q_1)$ (mm)	-39,9	-39,9	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	63	Rév. : D

6.5 CASE S03b: Steel beam with welded profile and MV interaction

6.5.1 Description

This test deals with a steel beam with welded profile and a MV interaction.

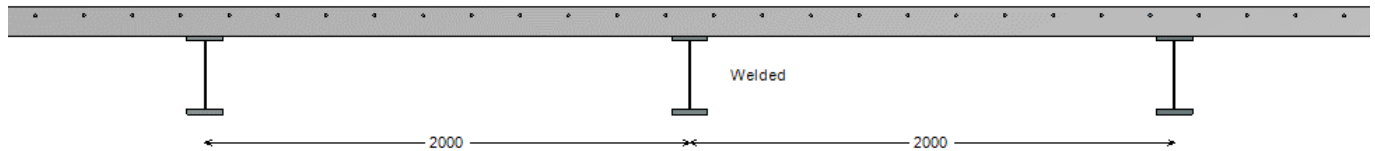


Figure 12 : Transversal view

6.5.2 Parameters

The case S03b is identical to Case S03, except that the cantilever is defined on the right end of the beam.

6.5.3 Verification of results

CASE S03B RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(Q_1) =$ (kN)	187,5	187,5	
Bending moment : $M(Q_1) =$ (kN.m)	-234,4	-234,4	

CASE S03B RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,850	0,850	
Resistance to shear force : Γ_V	0,703	0,703	
Resistance to shear buckling : Γ_{Vb}	-	-	
Resistance to MV interaction : Γ_{MV}	0,869	0,869	
Critical moment to lateral torsional buckling : $M_{cr,max,ULS-1}$ (kN.m)	632,5	647,8	+2,4%
Resistance to lateral torsional buckling : $\Gamma_{LT} =$	1,27	1,26	
Minimum design size of welds upper flange: a_w (mm)	1,40	1,40	
Minimum design size of welds lower flange: a_w (mm)	1,47	1,47	

CASE S03B RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection at the end of the cantilever : $\delta(Q_1)$ (mm)	69,7	69,7	
Deflection at mid span : $\delta(Q_1)$ (mm)	-39,9	-39,9	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	64	Rév. : D

6.6 CASE S04: Steel beam with welded profile and interaction between bending and shear buckling

6.6.1 Description

This test deals with a steel beam with welded profile and a MV interaction.

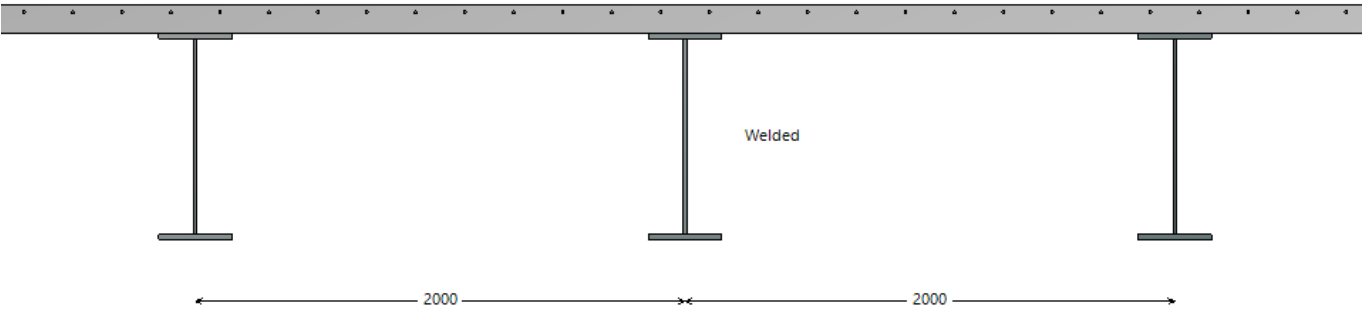


Figure 13 : Transversal view

6.6.2 Parameters

Definition of the beam:

Single span with a cantilever span on the left

Span length: 9 m

Cantilever length: 2,5 m

Definition of the profile:

Welded profile

Upper flange: 150x18

Web: 280x5

Lower flange: 150x22

Steel grade: S460 EC3

Definition of loads:

live load Q_l: distributed load on the cantilever span: 120 kN/m

Lateral restraints:

2 lateral restraints of both flange at 1/3 and 2/3 of the span length

Partial factors and national factors:

$$\gamma_{M0} = 1,0$$

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	65	Rév. : D

$$\gamma_{M1} = 1,0$$

$$\gamma_{M2} = 1,25$$

$$\eta = 1,2$$

Combination of loads:

$$\text{ULS-01:} \quad Q_1$$

$$\text{SLS:} \quad Q_1$$

Design parameters:

Standard: Eurocodes first generation

Working draft

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	66	Rév. : D

6.6.3 Verification of results

CASE S04 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	74	74	
Shear area : $A_v =$ (cm ²)	16,8	16,8	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	14 329	14 329	
Location of the elastic neutral axis: $z_{y,ENA} =$ (mm)	171,0	171	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	1 125,3	1 125	
Mass by length unit : $m_{lin} =$ (kg/ml)	58,1	58,1	
Mass : $m =$ (kg)	668	668	
Total paint surface : $S =$ (cm ²)	14,15	14,15	
Paint surface (without upper face) : $S =$ (cm ²)	12,42	12,42	
Nominal yield strength flanges : $f_y =$ (MPa)	460	460	
Nominal yield strength web : $f_y =$ (MPa)	460	460	
Axial resistance : $N_{t,Rd} =$ (kN)	3 404	3 404	
Shear resistance : $V_{pl,Rd} =$ (kN)	446,2	446,2	
Plastic bending resistance (y-y axis) : $M_{pl,Rd} =$ (kNm)	451	451	
Location of the plastic neutral axis: $z_{y,PNA} =$ (mm)	218	218	
Elastic bending resistance (y-y axis) : $M_{el,Rd} =$ (kNm)	386	386	
Section class for bending :	3		
Section class for compression :	4	4	
Shear buckling resistance : $h_w/t_w =$	56,0	56,0	
Shear buckling resistance : $72\varepsilon/\eta =$	42,9	42,9	
Verification of web buckling required?	Yes	Yes	

CASE S04 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(Q_1) =$ (kN)	300	300	
Bending moment : $M(Q_1) =$ (kN.m)	-375	-375	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	67	Rév. : D

CASE S04 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,973	0,973	
Resistance to shear force : Γ_V	0,672	0,672	
Resistance to shear buckling : Γ_{Vb}	0,881	0,881	
Resistance to MV interaction : Γ_{MV}	0,933	0,933	

CASE S04 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection at the end of the cantilever : $\delta(Q_1)$ (mm)	112,9	112,9	
Deflection at mid span : $\delta(Q_1)$ (mm)	-64,8	-64,8	

6.7 CASE S04b: Steel beam with welded profile and interaction between bending and shear buckling

6.7.1 Description

This test deals with a steel beam with welded profile and a MV interaction.

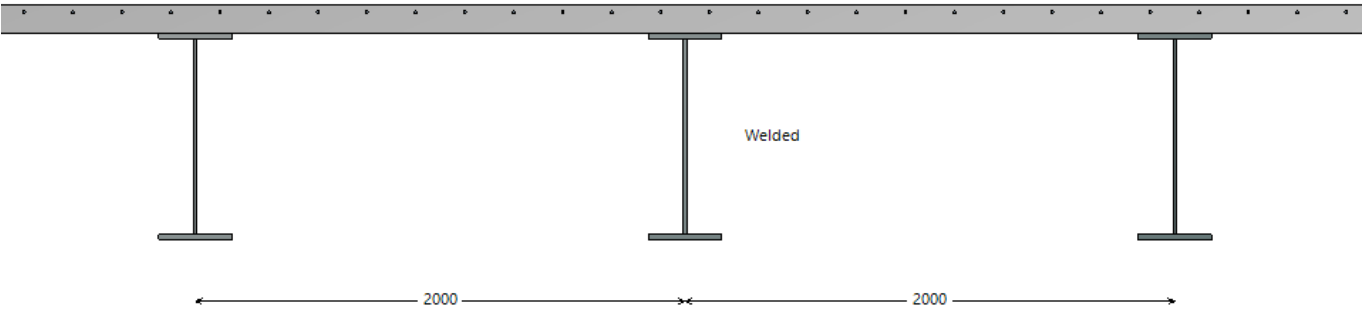


Figure 14 : Transversal view

6.7.2 Parameters

The case S04b is identical to Case S04, except that the cantilever is defined on the right end of the beam.

6.7.3 Verification of results

CASE S04B RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(Q_1)$ = (kN)	300	300	
Bending moment : $M(Q_1)$ = (kN.m)	-375	-375	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	69	Rév. : D

CASE S04 RESULTS: ULS	Reference	ABCPMX	Dev.
Resistance to bending moment : Γ_M	0,973	0,973	
Resistance to shear force : Γ_V	0,672	0,672	
Resistance to shear buckling : Γ_{Vb}	0,881	0,881	
Resistance to MV interaction : Γ_{MV}	0,933	0,933	

CASE S04 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection at the end of the cantilever: $\delta(Q_1)$ (mm)	112,9	112,9	
Deflection at mid span: $\delta(Q_1)$ (mm)	-64,8	-64,8	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	70	Rév. : D

7 TESTS CASES – FIRE DESIGN

7.1 CASE F01: Steel beam with unprotected hot rolled section

7.1.1 Description

This test deals with an unprotected steel beam.

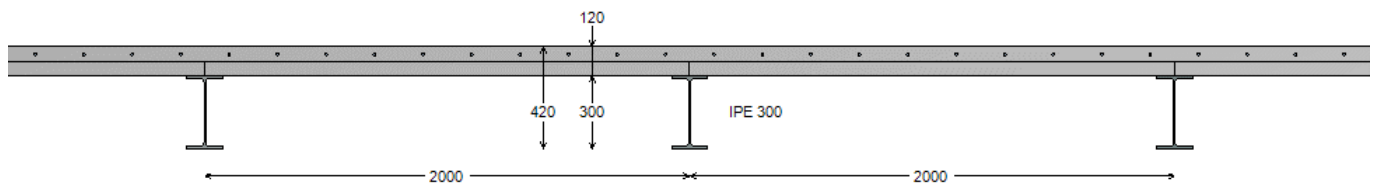


Figure 15 : Transversal view

7.1.2 Parameters

Definition of the beam:

Total length: $L = 9 \text{ m}$

Definition of the profile:

Hot rolled profile

Profile: IPE 300

Steel grade: S275 JR

Lateral restraints:

The beam is laterally restrained at mid-span (both flanges)

Definition of loads:

Live load Q_1 : distributed load 5 kN/m

Combination of loads:

Fire ULS: $1 \times Q_1$

Design parameters:

Standard: Eurocodes first generation

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	71	Rév. : D

Fire parameters:

Steel surface: Not galvanized, not protected
Time increment: $\Delta t = 5 \text{ s}$

7.1.3 Verification of results

CASE F01 RESULTS: HEATING OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
Temperature at 30 mn : $\theta =$ (°C)	797,9	798	
Temperature at 60 mn : $\theta =$ (°C)	939,6	940	
Temperature at 90 mn : $\theta =$ (°C)	1002,7	1003	
Temperature at 120 mn : $\theta =$ (°C)	1046,8	1047	
Temperature at 180 mn : $\theta =$ (°C)	1108,5	1109	
Temperature at 240 mn : $\theta =$ (°C)	1151,9	1152	

CASE F01 RESULTS: RESISTANCE OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
R30 Results			
Reduction factor : $k_{y,\theta} =$	0,113	0,112	
Reduction factor : $k_{E,\theta} =$	0,091	0,091	
Bending resistance criterion : $\Gamma_{M\theta} =$	2,60	2,62	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,490	0,493	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	6,43	6,46	
R60 Results			
Reduction factor : $k_{y,\theta} =$	0,052	0,052	
Reduction factor : $k_{E,\theta} =$	0,059	0,059	
Bending resistance criterion : $\Gamma_{M\theta} =$	5,63	5,63	
Shear resistance criterion : $\Gamma_{V\theta} =$	1,060	1,060	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	11,66	11,66	
R90 Results			
Reduction factor : $k_{y,\theta} =$	0,039	0,039	
Reduction factor : $k_{E,\theta} =$	0,044	0,044	
Bending resistance criterion : $\Gamma_{M\theta} =$	7,43	7,43	
Shear resistance criterion : $\Gamma_{V\theta} =$	1,400	1,40	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	15,39	15,39	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	72	Rév. : D

7.2 CASE F01b: Steel beam with unprotected galvanized hot rolled section

7.2.1 Description

This test deals with an unprotected galvanized steel beam.

7.2.2 Parameters

The case F01b is identical to Case F01, except that the steel profile is galvanized.

7.2.3 Verification of results

CASE F01b RESULTS: HEATING OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
Temperature at 30 mn : $\theta =$ (°C)	787,9	789	
Temperature at 60 mn : $\theta =$ (°C)	939,6	940	
Temperature at 90 mn : $\theta =$ (°C)	1002,7	1003	

CASE F01b RESULTS: RESISTANCE OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
R30 Results			
Reduction factor : $k_{y,\theta} =$	0,124	0,124	
Reduction factor : $k_{E,\theta} =$	0,095	0,095	
Bending resistance criterion : $\Gamma_{M\theta} =$	2,36	2,37	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,443	0,447	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	6,02	6,05	
R60 Results			
Reduction factor : $k_{y,\theta} =$	0,052	0,052	
Reduction factor : $k_{E,\theta} =$	0,059	0,059	
Bending resistance criterion : $\Gamma_{M\theta} =$	5,63	5,63	
Shear resistance criterion : $\Gamma_{V\theta} =$	1,060	1,060	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	11,66	11,66	
R90 Results			
Reduction factor : $k_{y,\theta} =$	0,039	0,039	
Reduction factor : $k_{E,\theta} =$	0,044	0,044	
Bending resistance criterion : $\Gamma_{M\theta} =$	7,43	7,43	
Shear resistance criterion : $\Gamma_{V\theta} =$	1,400	1,40	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	15,39	15,39	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	73	Rév. : D

7.3 CASE F02: Steel beam with protected welded section (spray)

7.3.1 Description

This test deals with a protected steel beam.

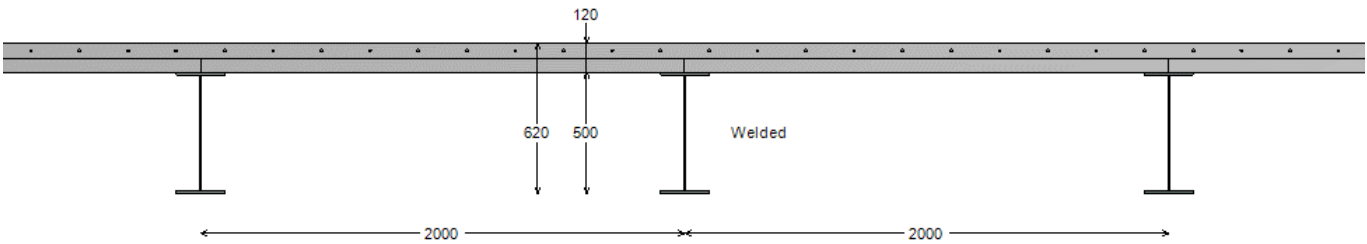


Figure 16 : Transversal view

7.3.2 Parameters

Definition of the beam:

Total length: $L = 12\text{ m}$

Definition of the profile:

Welded profile
Upper flange: 200x12
Web: 476x8
Lower flange: 200x12
Steel grade: S355 EC3

Lateral restraints:

2 lateral restraints of both flange at 1/3 and 2/3 of the span length

Definition of loads:

live load Q1: distributed load 5 kN/m

Combination of loads:

Fire ULS: 1 x Q1

Design parameters:

Standard: Eurocodes first generation

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	74	Rév. : D

Fire parameters:

Steel surface: protected
Protection: Spray by mineral fibers, thickness 20 mm
Time increment: $\Delta t = 5 \text{ s}$

7.3.3 Verification of results

CASE F02 RESULTS: HEATING OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
Temperature at 30 mn : $\theta =$ (°C)	256	257	
Temperature at 60 mn : $\theta =$ (°C)	463	463	
Temperature at 90 mn : $\theta =$ (°C)	608	608	
Temperature at 120 mn : $\theta =$ (°C)	709	709	
Temperature at 180 mn : $\theta =$ (°C)	829	829	
Temperature at 240 mn : $\theta =$ (°C)	983	983	

CASE F02 RESULTS: RESISTANCE OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
R30 Results			
Reduction factor : $k_{y,\theta} =$	1	1	
Reduction factor : $k_{E,\theta} =$	0,844	0,843	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,177	0,177	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,038	0,038	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,050	0,050	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,335	0,335	
R60 Results			
Reduction factor : $k_{y,\theta} =$	0,862	0,861	
Reduction factor : $k_{E,\theta} =$	0,637	0,637	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,205	0,206	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,0446	0,045	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,06218	0,062	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,413	0,414	
R90 Results			
Reduction factor : $k_{y,\theta} =$	0,451	0,450	
Reduction factor : $k_{E,\theta} =$	0,295	0,295	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,393	0,394	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,0853	0,085	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,126	0,127	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,841	0,843	

7.4 CASE F02b: Steel beam with protected welded section (intumescent paint)

7.4.1 Description

This test deals with a protected steel beam.

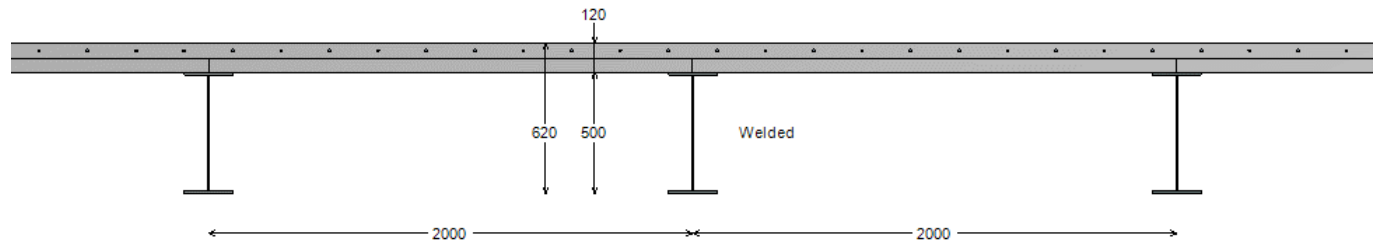


Figure 17 : Transversal view

7.4.2 Parameters

The case F02b is identical to Case F02, except that the steel profile is protected by an intumescent painting, with following parameters:

Fire parameters:

Steel surface: protected
Protection: Intumescent paint, thickness 5 mm, thermal conductivity $\lambda_p = 0,01 \text{ W/mK}$

7.4.3 Verification of results

CASE F02B RESULTS: HEATING OF THE STEEL PROFILE		Reference	ABCPMX	Dev.
Temperature at 30 mn : $\theta =$	(°C)	131	131	
Temperature at 60 mn : $\theta =$	(°C)	245	246	
Temperature at 90 mn : $\theta =$	(°C)	347	347	
Temperature at 120 mn : $\theta =$	(°C)	436	436	
Temperature at 180 mn : $\theta =$	(°C)	579	579	
Temperature at 240 mn : $\theta =$	(°C)	684	684	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	76	Rév. : D

CASE F02B RESULTS: RESISTANCE OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
R30 Results			
Reduction factor : $k_{y,\theta} =$	1	1	
Reduction factor : $k_{E,\theta} =$	0,969	0,969	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,177	0,177	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,038	0,038	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,047	0,047	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,316	0,316	
R60 Results			
Reduction factor : $k_{y,\theta} =$	1	1	
Reduction factor : $k_{E,\theta} =$	0,855	0,854	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,177	0,177	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,038	0,038	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,050	0,050	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,333	0,333	
R90 Results			
Reduction factor : $k_{y,\theta} =$	1	1	
Reduction factor : $k_{E,\theta} =$	0,753	0,753	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,177	0,177	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,038	0,038	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,053	0,053	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,353	0,353	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	77	Rév. : D

7.5 CASE F02c: Steel beam with protected welded section (plaster boards)

7.5.1 Description

This test deals with a protected steel beam.

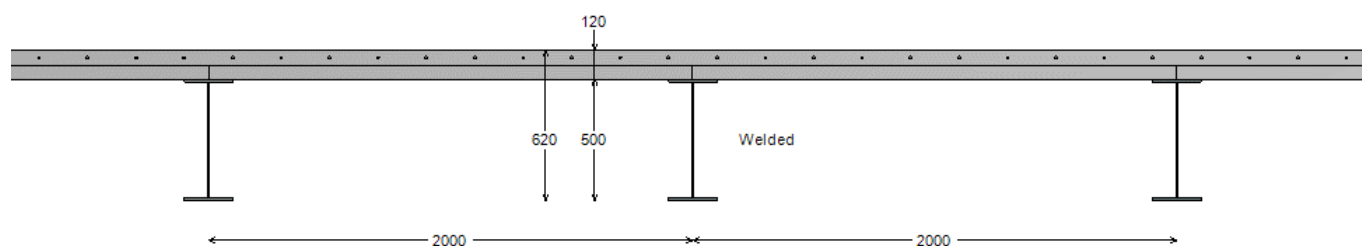


Figure 18 : Transversal view

7.5.2 Parameters

The case F02c is identical to Case F02, except that the steel profile is protected by plaster boards, with following parameters:

Fire parameters:

Steel surface: protected
Protection: Plaster boards, thickness 25 mm

7.5.3 Verification of results

CASE F02C RESULTS: HEATING OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
Temperature at 30 mn : $\theta =$ (°C)	187,2	188	
Temperature at 60 mn : $\theta =$ (°C)	381,0	382	
Temperature at 90 mn : $\theta =$ (°C)	532,5	533	
Temperature at 120 mn : $\theta =$ (°C)	646,5	647	
Temperature at 180 mn : $\theta =$ (°C)	765,7	765	
Temperature at 240 mn : $\theta =$ (°C)	918,8	919	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	78	Rév. : D

CASE F02C RESULTS: RESISTANCE OF THE STEEL PROFILE	Reference	ABCPMX	Dev.
R30 Results			
Reduction factor : $k_{y,\theta} =$	1	1	
Reduction factor : $k_{E,\theta} =$	0,913	0,912	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,177	0,177	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,038	0,038	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,0483	0,048	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,324	0,324	
R60 Results			
Reduction factor : $k_{y,\theta} =$	1	1	
Reduction factor : $k_{E,\theta} =$	0,719	0,718	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,177	0,177	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,038	0,038	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,0544	0,054	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,362	0,362	
R90 Results			
Reduction factor : $k_{y,\theta} =$	0,679	0,678	
Reduction factor : $k_{E,\theta} =$	0,506	0,505	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,261	0,261	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,056	0,057	
Shear buckling resistance criterion : $\Gamma_{Vb\theta} =$	0,0786	0,079	
LTBuckling resistance criterion: $\Gamma_{LT\theta} =$	0,523	0,524	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	79	Rév. : D

7.6 CASE F03: Composite beam with unprotected hot rolled profile

7.6.1 Description

This test deals with an unprotected composite beam.

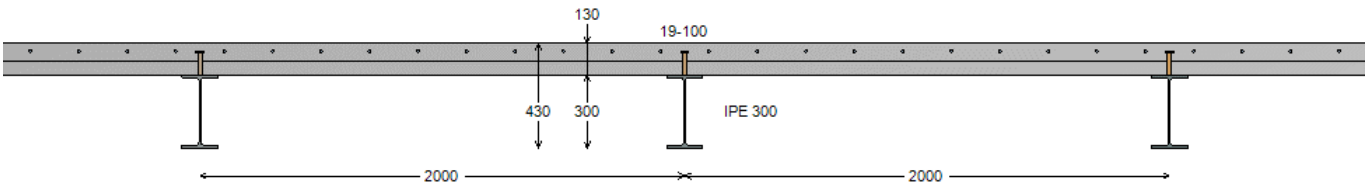


Figure 19 : Transversal view

7.6.2 Parameters

Definition of the beam:

Total length: $L = 9\text{ m}$

Definition of the profile:

Hot rolled profile
Profile: IPE 300
Steel grade: S275 JR

Definition of the slab:

Composite slab
Total depth: $t_d = 130\text{ mm}$
Perpendicular ribs
Steel profiled sheeting: Cofraplus 60, thickness 0,75 mm
Prepunched sheeting

Definition of the concrete:

Normal concrete
Concrete class: C25/30
Density: $\rho_c = 2500\text{ kg/m}^3$

Definition of the connection:

Studs: 19x100
One stud in each rib
Unpropped beam during construction

Definition of loads:

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	80	Rév. : D

live load Q1: distributed load 5 kN/m

Combination of loads:

Fire ULS: 1 x Q1

Design parameters:

Standard: Eurocodes first generation

Fire parameters:

Steel surface: unprotected

Time increment: $\Delta t = 5 \text{ s}$

7.6.3 Verification of results

CASE F03 RESULTS: HEATING OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Temperature of the upper flange : $\theta =$ (°C)	734,1	734	
Reduction factor : $k_{y\theta} =$	0,189	0,189	
Temperature of the web : $\theta =$ (°C)	824	825	
Reduction factor : $k_{y\theta} =$	0,098	0,098	
Temperature of the lower flange : $\theta =$ (°C)	802	803	
Reduction factor : $k_{y\theta} =$	0,109	0,109	
Temperature of the lower fibre of the slab: $\theta =$ (°C)	535	535	
Reduction factor : $k_{c\theta} =$	0,548	0,548	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	60	60	
Reduction factor : $k_{c\theta} =$	1,0	1,0	
Temperature of the stud shank : $\theta =$ (°C)	587	587	
Reduction factor : $k_{u\theta} =$	0,510	0,509	
Temperature the concrete around the stud: $\theta =$ (°C)	294	294	
Reduction factor : $k_{c\theta} =$	0,856	0,856	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	81	Rév. : D

CASE F03 RESULTS: RESISTANCE OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Shear resistance of the connector : P_{Rd} = (kN)	29,6	29,6	
Bending resistance criterion : $\Gamma_{M\theta}$ =	1,061	1,064	
Shear resistance criterion : $\Gamma_{V\theta}$ =	0,563	0,565	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	82	Rév. : D

7.7 CASE F03b: Composite beam with galvanised hot rolled profile

7.7.1 Description

This test deals with an unprotected composite beam. The steel profile is galvanised.

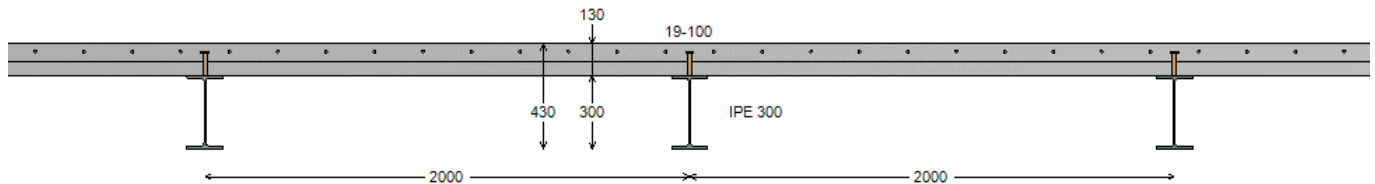


Figure 20 : Transversal view

7.7.2 Parameters

The case F03b is identical to Case F03, except that the steel profile is galvanized.

7.7.3 Verification of results

CASE F03B RESULTS: HEATING OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Temperature of the upper flange : $\theta =$ (°C)	709	711	
Reduction factor : $k_{y\theta} =$	0,219	0,217	
Temperature of the web : $\theta =$ (°C)	823	823	
Reduction factor : $k_{y\theta} =$	0,099	0,098	
Temperature of the lower flange : $\theta =$ (°C)	794	795	
Reduction factor : $k_{y\theta} =$	0,117	0,116	
Temperature of the lower fibre of the slab: $\theta =$ (°C)	535	535	
Reduction factor : $k_{c\theta} =$	0,548	0,548	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	60	60	
Reduction factor : $k_{c\theta} =$	1,0	1,0	
Temperature of the stud shank : $\theta =$ (°C)	567	569	
Reduction factor : $k_{u\theta} =$	0,571	0,567	
Temperature the concrete around the stud: $\theta =$ (°C)	284	284	
Reduction factor : $k_{c\theta} =$	0,866	0,866	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	83	Rév. : D

CASE F03B RESULTS: RESISTANCE OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Shear resistance of the connector : P_{Rd} = (kN)	33,21	33,0	
Bending resistance criterion : $\Gamma_{M\theta}$ =	0,990	0,997	
Shear resistance criterion : $\Gamma_{V\theta}$ =	0,560	0,562	

7.8 CASE F04: Composite beam with protected welded profile

7.8.1 Description

This test deals with a protected composite beam.

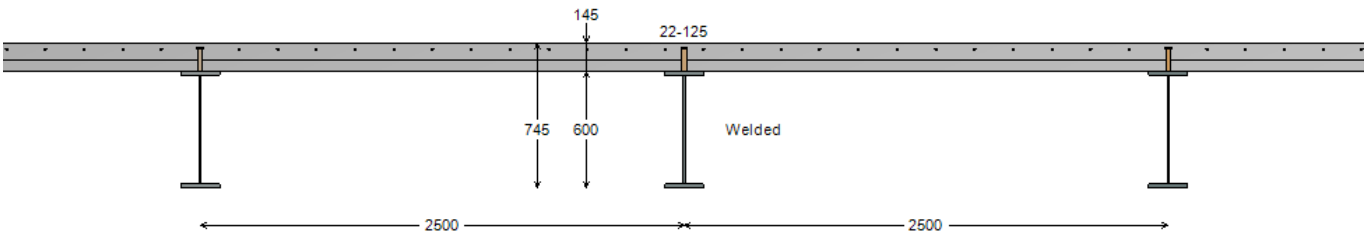


Figure 21 : Transversal view

7.8.2 Parameters

Definition of the beam:

Total length: $L = 14\text{ m}$

Definition of the profile:

Welded profile
Upper flange: 200x20
Web: 576x10
Lower flange: 200x20
Steel grade: S355 JR

Definition of the slab:

Composite slab
Total depth: $t_d = 145\text{ mm}$
Perpendicular ribs
Steel profiled sheeting: Cofraplus 60, thickness 0,75 mm
Prepunched sheeting

Definition of the concrete:

Normal concrete
Concrete class: C30/35
Density: $\rho_c = 2500\text{ kg/m}^3$

Definition of the connection:

Studs: 22x125
Two studs in each rib
Unpropped beam during construction

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	85	Rév. : D

Definition of loads:

live load Q1: distributed load 5 kN/m

Combination of loads:

Fire ULS: 1 x Q1

Design parameters:

Standard: Eurocodes first generation

Fire parameters:

Steel surface: protected

Protection: Spray by perlite, thickness 25 mm

Time increment: $\Delta t = 5 \text{ s}$

Working draft

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	86	Rév. : D

7.8.3 Verification of results

CASE F04 RESULTS: HEATING OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Temperature of the upper flange : $\theta =$ (°C)	92,1	92,4	
Reduction factor : $k_{y\theta} =$	1,0	1,0	
Temperature of the web : $\theta =$ (°C)	194,3	195	
Reduction factor : $k_{y\theta} =$	1,0	1,0	
Temperature of the lower flange : $\theta =$ (°C)	136,2	137	
Reduction factor : $k_{y\theta} =$	1,0	1,0	
Temperature of the lower fibre of the slab: $\theta =$ (°C)	535	535	
Reduction factor : $k_{c\theta} =$	0,548	0,548	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	60	60	
Reduction factor : $k_{c\theta} =$	1,0	1,0	
Temperature of the stud shank : $\theta =$ (°C)	74	73,9	
Reduction factor : $k_{u\theta} =$	1,25	1,25	
Temperature the concrete around the stud: $\theta =$ (°C)	37	37	
Reduction factor : $k_{c\theta} =$	1,0	1,0	
R180 Results			
Temperature of the upper flange : $\theta =$ (°C)	479,0	479	
Reduction factor : $k_{y\theta} =$	0,826	0,826	
Temperature of the web : $\theta =$ (°C)	750,6	750	
Reduction factor : $k_{y\theta} =$	0,169	0,170	
Temperature of the lower flange : $\theta =$ (°C)	640,3	644	
Reduction factor : $k_{y\theta} =$	0,373	0,365	
Temperature of the lower fibre of the slab: $\theta =$ (°C)	1110	1110	
Reduction factor : $k_{c\theta} =$	0,009	0,009	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	260	260	
Reduction factor : $k_{c\theta} =$	0,890	0,890	
Temperature of the stud shank : $\theta =$ (°C)	383	383	
Reduction factor : $k_{u\theta} =$	1,042	1,042	
Temperature the concrete around the stud: $\theta =$ (°C)	192	192	
Reduction factor : $k_{c\theta} =$	0,954	0,954	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	87	Rév. : D

CASE F04 RESULTS: RESISTANCE OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Shear resistance of the connector : P_{Rd} = (kN)	82,11	82,1	
Bending resistance criterion : $\Gamma_{M\theta}$ =	0,0622	0,062	
Shear resistance criterion : $\Gamma_{V\theta}$ =	0,03049	0,030	
R180 Results			
Shear resistance of the connector : P_{Rd} = (kN)	68,44	68,4	
Bending resistance criterion : $\Gamma_{M\theta}$ =	0,182	0,183	
Shear resistance criterion : $\Gamma_{V\theta}$ =	0,180	0,180	

7.9 CASE F04b: Composite beam with protected welded profile (board)

7.9.1 Description

This test deals with a protected composite beam.

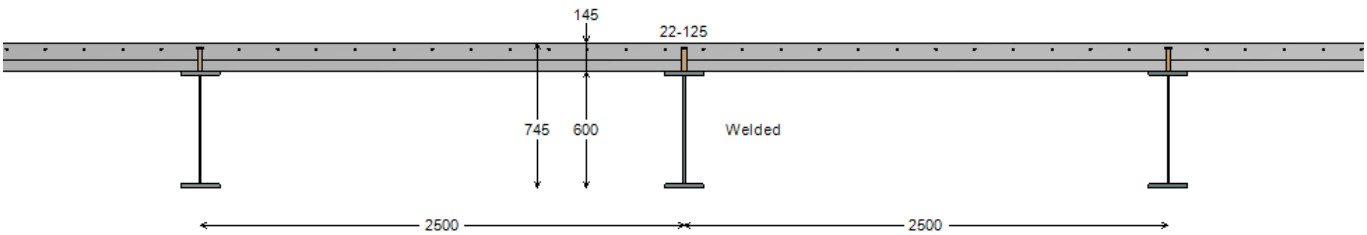


Figure 22 : Transversal view

7.9.2 Parameters

The case F04b is identical to Case F04, except that the steel profile is protected by an intumescent painting, with following parameters:

Fire parameters:

- Steel surface: protected
- Protection: Silicate fiber board, thickness 30 mm,

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	89	Rév. : D

7.9.3 Verification of results

CASE F04B RESULTS: HEATING OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Temperature of the steel profile : $\theta =$ (°C)	123,5	124	
Reduction factor : $k_{y\theta} =$	1,0	1,0	
Temperature of the lower fibre of the slab: $\theta =$ (°C)	535	535	
Reduction factor : $k_{c\theta} =$	0,548	0,548	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	60	60	
Reduction factor : $k_{c\theta} =$	1,0	1,0	
Temperature of the stud shank : $\theta =$ (°C)	99	99,1	
Reduction factor : $k_{u\theta} =$	1,25	1,25	
Temperature the concrete around the stud: $\theta =$ (°C)	49	49,5	
Reduction factor : $k_{c\theta} =$	1,0	1,0	
R180 Results			
Temperature of the steel profile : $\theta =$ (°C)	622	622	
Reduction factor : $k_{y\theta} =$	0,417	0,417	
Temperature of the lower fibre of the slab: $\theta =$ (°C)	1110	1110	
Reduction factor : $k_{c\theta} =$	0,009	0,009	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	260	260	
Reduction factor : $k_{c\theta} =$	0,890	0,890	
Temperature of the stud shank : $\theta =$ (°C)	498	498	
Reduction factor : $k_{u\theta} =$	0,785	0,785	
Temperature the concrete around the stud: $\theta =$ (°C)	249	249	
Reduction factor : $k_{c\theta} =$	0,901	0,901	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	90	Rév. : D

CASE F04B RESULTS: RESISTANCE OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Shear resistance of the connector : $P_{Rd} =$ (kN)	82,1	82,1	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,062	0,062	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,030	0,030	
Shear resistance of the connector : $P_{Rd} =$ (kN)	51,6	51,6	
Bending resistance criterion : $\Gamma_{M\theta} =$	0,1431	0,144	
Shear resistance criterion : $\Gamma_{V\theta} =$	0,073	0,073	

7.10 CASE F05: Composite beam with protected welded profile and solid slab

7.10.1 Description

This test deals with a protected composite beam and a solid slab.

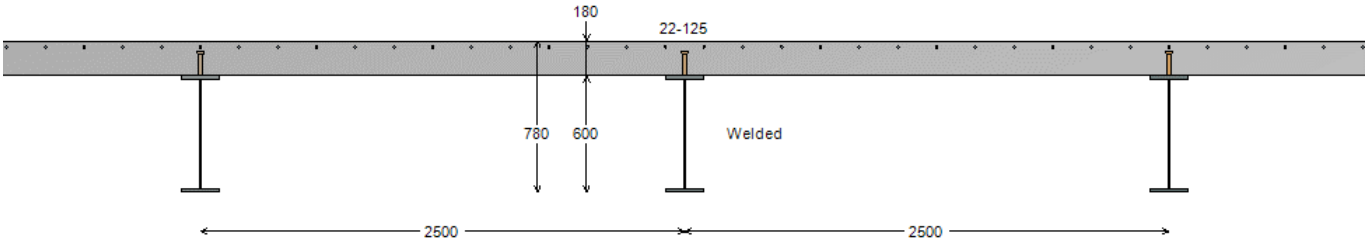


Figure 23 : Transversal view

7.10.2 Parameters

Definition of the beam:

Total length: $L = 14\text{ m}$

Definition of the profile:

Welded profile
Upper flange: 200x20
Web: 560x10
Lower flange: 200x20
Steel grade: S355 JR

Definition of the slab:

Solid slab
Total depth: $t_d = 180\text{ mm}$

Definition of the concrete:

Normal concrete
Concrete class: C30/37
Density: $\rho_c = 2300\text{ kg/m}^3$

Definition of the connection:

Studs: 22x125
One stud every 200 mm
Unpropped beam during construction

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	92	Rév. : D

Definition of loads:

live load Q_l: distributed load 5 kN/m

Combination of loads:

Fire ULS: 1 x Q_l

Design parameters:

Standard: Eurocodes first generation

Fire parameters:

Steel surface: protected

Protection: Spray by mineral fibres, thickness 20 mm

Time increment: $\Delta t = 5$ s

Heating of the slab obtained by numerical simulation

Variable density of concrete

French NA is not applied

Moisture content $u = 0\%$

Emissivity of concrete $\varepsilon_c = 0,7$

7.10.3 Verification of results

Reference values have been obtained by a numerical FEM calculation with ANSYS

CASE F05 RESULTS: HEATING OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Temperature of the lower fibre of the slab: $\theta =$ (°C)	603,0	604	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	20,7	20,4	
R180 Results			
Temperature of the lower fibre of the slab: $\theta =$ (°C)	1006	1006	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	144,4	144	

7.11 CASE F05B: Composite beam with protected welded profile and solid slab

7.11.1 Parameters

The case F05b is identical to Case F05, except the following parameters:

Fire parameters:

Steel surface: protected

Protection: Spray by mineral fibres, thickness 20 mm

Time increment: $\Delta t = 5$ s

Heating of the slab obtained by numerical simulation

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	93	Rév. : D

Constant density of concrete

French NA is applied

Moisture content $u = 0 \%$

Emissivity of concrete $\varepsilon_c = 0,7$

7.11.2 Verification of results

Reference values have been obtained by a numerical FEM calculation with ANSYS

CASE F05B RESULTS: HEATING OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Temperature of the lower fibre of the slab: $\theta =$ (°C)	602	603	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	20,5	20,5	
R180 Results			
Temperature of the lower fibre of the slab: $\theta =$ (°C)	1006	1006	
Temperature of the upper fibre of the slab: $\theta =$ (°C)	117,6	118	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	94	Rév. : D

7.12 CASE F06: Composite beam partially encased profile

7.12.1 Description

This test deals with composite beam with a partially encased steel profile and a composite slab.
This test corresponds to an example published in the review Construction Métallique, n°1999-1.

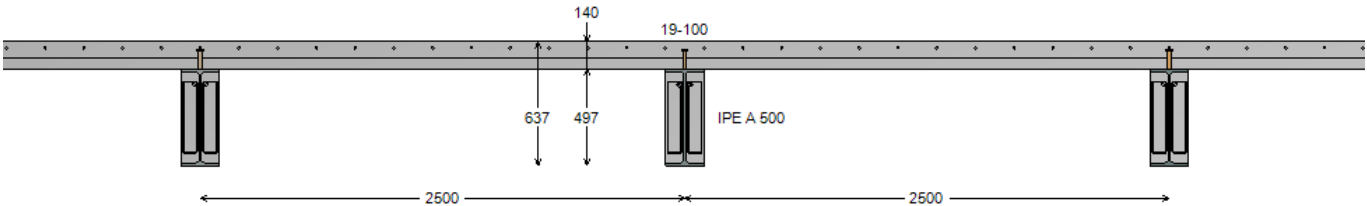


Figure 24 : Transversal view

7.12.2 Parameters

Definition of the beam:

Total length: $L = 12.5\text{ m}$

Definition of the profile:

IPE A 500
Steel grade: S355 JR

Definition of the slab:

Composite slab
Total depth: $t_d = 140\text{ mm}$
Cofraplus 60

Definition of the concrete (slab):

Normal concrete
Concrete class: C30/37
Density: $\rho_c = 2500\text{ kg/m}^3$

Definition of the connection:

Studs: 19x100
One stud every trough
Unpropped beam during construction

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	95	Rév. : D

Definition of the concrete encasement:

Definition of partial encasement

Dimensions

Width

$b_c = 1$

x b_f

$b_c = 200$

mm

Stirrups

Type

Closed stirrups

Diameter

$\varphi_e = 6$

mm

Coverage

$u_y = 10$

mm

$u_z = 49$

mm

Concrete

Class

C30/37

$f_{ck} = 30$

MPa

$E_{cm} = 32.837$

GPa

$\rho_{ce} = 2500$

kg/m3

Steel of the reinforcement

Class

B500

$f_{sk} = 500$

MPa

Longitudinal reinforcement

External

Middle

Internal

Number

Single

None

Single

Diameter

8

8

12

mm

Active rebars

☒
☐
☒

Position

$z_{s,inf} = 422$

mm

Lower layer

$A_{s,inf} = 163.363$

mm2

Definition of loads:

live load Q1: distributed load 20 kN/m

Combination of loads:

Fire ULS: 1 x Q1

Design parameters:

Standard: Eurocodes first generation

Fire parameters:

Steel surface: unprotected

Root fillet not considered

Partial factors in fire situations:

$\gamma_{M,fi} = 1$

$\gamma_{C,fi} = 1,3$

$\gamma_{s,fi} = 1$

$\gamma_{V,fi} = 1$

7.12.3 Verification of results

CASE F06 RESULTS: ULTIMATE LIMIT STATE		Reference	ABCPMX	Dev.
R90 Results				
Bending resistance criterion:	$\Gamma_{M\theta} =$	0,624	0,634	+1,6 %
Shear resistance criterion:	$\Gamma_{V\theta} =$	0,167	0,167	

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	97	Rév. : D

Definition of the profile:

Welded profile
Upper flange: 120x15
Web: 270x8
Lower flange: 120x15
Steel grade: S355 M/ML

Definition of the slab:

Composite slab
Total depth: $t_d = 140 \text{ mm}$
Cofraplus 60

Definition of the concrete:

Normal concrete
Concrete class: C30/37
Density: $\rho_c = 2500 \text{ kg/m}^3$

Definition of the connection:

Studs: 19x100
One stud every trough
Unpropped beam during construction

Definition of loads:

live load Q1: distributed load 20 kN/m

Combination of loads:

Fire ULS: 1 x Q1

Design parameters:

Standard: Eurocodes first generation

Fire parameters:

Steel surface: protected
Protection: Spray by perlite, thickness 25 mm
The voids between the upper flange and the steel profiled sheeting are not filled

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	98	Rév. : D

7.13.3 Verification of results

CASE F07 RESULTS: HEATING OF THE BEAM	Reference	ABCPMX	Dev.
R30 Results			
Temperature of the upper flange: $\theta =$ (°C)	264	265	
Temperature of the web: $\theta =$ (°C)	219	219	
Temperature of the lower flange: $\theta =$ (°C)	165	165	
R60 Results			
Temperature of the upper flange: $\theta =$ (°C)	506	507	
Temperature of the web: $\theta =$ (°C)	424	425	
Temperature of the lower flange: $\theta =$ (°C)	330	331	
R90 Results			
Temperature of the upper flange: $\theta =$ (°C)	673	673	
Temperature of the web: $\theta =$ (°C)	578	579	
Temperature of the lower flange: $\theta =$ (°C)	484	485	
R120 Results			
Temperature of the upper flange: $\theta =$ (°C)	777	753	
Temperature of the web: $\theta =$ (°C)	687	687	
Temperature of the lower flange: $\theta =$ (°C)	595	592	

8 TESTS CASES – SLIM FLOOR STEEL BEAMS

8.1 CASE SFS01: Steel beam with SFB steel section

8.1.1 Description

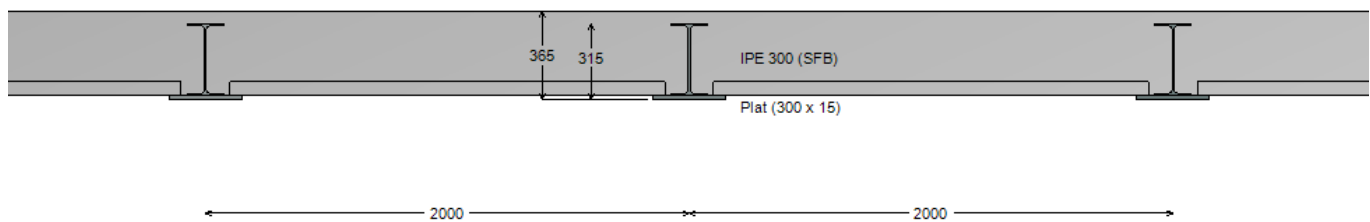
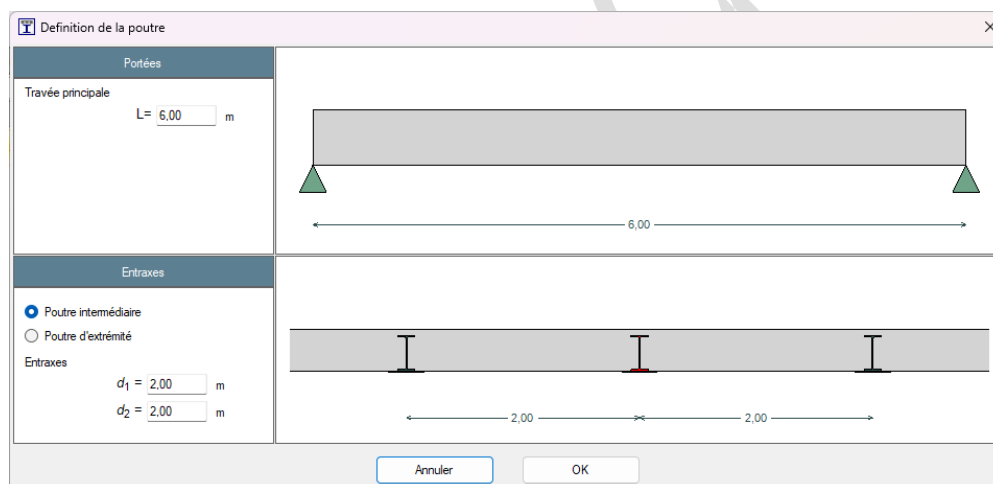


Figure 26 : Transversal view

8.1.2 Parameters

Definition of the beam:



Definition of the profile:

Hot rolled profile

Profile: IPE 300

Steel grade: S355 J0

Steel plate

Dimensions: 300x15

Steel grade: S355 J0

Définition de la dalle

Général	Bac acier
Type : Dalle mixte Épaisseur : ☒ $t_d = 350$ mm ☐ $t_c = 292$ mm	Nom : Cofraplus_60 T.00  Hauteur : $h_p = 58.00$ mm Modifier le bac acier
Béton ☐ Béton léger Classe : C20/25 $\rho_c = 2500$ kg/m³	
Acier des armatures Classe : B500 $f_{sk} = 500$ MPa	



Annuler
OK

G1: self-weight only
live load Q1: distributed load: 20 kN/m

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	101	Rév. : D

8.1.3 Verification of results

CASE S01 RESULTS: STEEL PROFILE		Reference	ABCPMX	Dev.
Cross-sectional area : $A =$	(cm ²)	98,81	98,81	
Shear area : $A_v =$	(cm ²)	25,68	25,68	
Second moment of area (y-y axis) : $I_y =$	(cm ⁴)	14443	14444	
Second moment of area (z-z axis) : $I_z =$	(cm ⁴)	3979	3978	
Mass by length unit : $m_{lin} =$	(kg/ml)	77,57	77,6	
Mass : $m =$	(kg)	465,4	465,4	
Nominal yield strength : $f_y =$	(MPa)	355	355	
Axial resistance : $N_{t,Rd} =$	(kN)	3507,8	3508	
Shear resistance : $V_{pl,Rd} =$	(kN)	526,4	526	
Plastic moment resistant : $M_{pl,Rd} =$	(kN.m)	298,1	298	
Elastic moment resistant : $M_{el,Rd} =$	(kN.m)	231,3	231	

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	102	Rév. : D

CASE SFS01 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	49,3	49,3	
Bending moment : $M(G_1) =$ (kN.m)	73,9	73,9	
Shear force : $V(Q_1\#1) =$ (kN)	60	60	
Bending moment : $M(Q_1\#1) =$ (kN.m)	90	90	
Shear force : $V_{Ed,ULS} =$ (kN)	156,5	156,5	
Bending moment : $M_{Ed,ULS} =$ (kN.m)	234,8	234,8	

CASE SFS01 RESULTS: ULS IN THE CENTRAL SPAN	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_{M,ULS} =$	1,015	1,015	
Resistance to shear force : $\Gamma_{V,ULS} =$	0,297	0,297	
Resistance to local bending : $\Gamma_{MV,ULS} =$	0,504	0,504	

CASE SFS01 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	9,14	9,14	
Deflection : $\delta(Q_1)$ (mm)	11,1	11,1	
Deflection : $\delta(SLS)$ (mm)	20,3	20,3	

8.2 CASE SFS01b: Steel beam with SFB steel section

8.2.1 Description

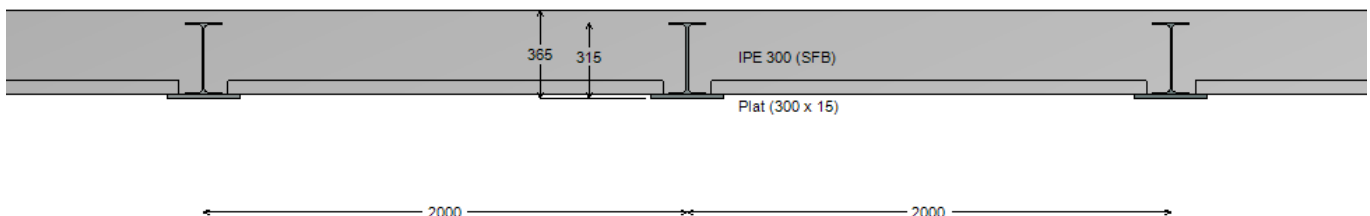


Figure 27 : Transversal view

8.2.2 Parameters

The case SFS01b is identical to Case SFS01, except that the steel profile, including the plate, is in S235 steel.

CTI

CM

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au service de
la construction
métallique

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Product reference: DRV/PMX/MV/001

Date:15/05/2025

Authors:

POM+GUD

Page:103

Rév. :D

8.2.3 Verification of results

CASE SFS01B RESULTS: ULS IN THE CENTRAL SPAN	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_{M,ULS} =$	1,196	1,192	
Resistance to shear force : $\Gamma_{V,ULS} =$	0,449	0,449	
Resistance to local bending : $\Gamma_{MV,ULS} =$	0,274	0,274	

8.3 CASE SFS02: Steel beam with IFB-A steel section

8.3.1 Description

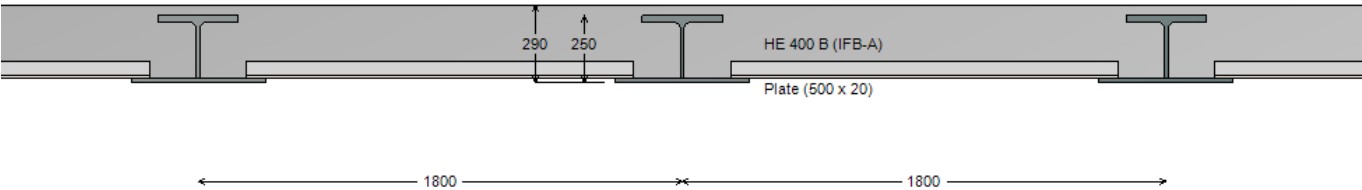
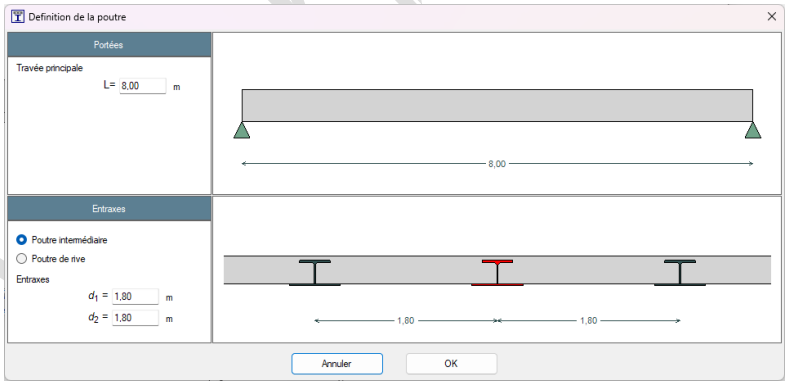


Figure 28 : Transversal view

8.3.2 Parameters

Definition of the beam:



Definition of the profile:

Profile: HEB 400
Steel grade: S355 J0

Steel plate

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	104	Rév. : D

Dimensions:

500x20

Steel grade:

S355 JO

Definition of the slab:

Definition of the slab

Main

Rebars

General

Type

Partially prefabricated slab

Thickness

t_g = 270 mm

Prefab. part

t_{pre} = 60 mm

Concrete part

t_c = 50 mm

Concrete

☐ Light concrete

Class

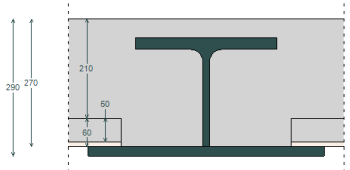
C20/25

2500 kg/m3

Masses

Mass per surface unit of the concrete, including the prefabricated part

675 kg/m2



Cancel

OK

Definition of loads:

G1:

self-weight only

live load Q1:

distributed load:

30 kN/m

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	105	Rév. : D

8.3.3 Verification of results

CASE SFS02 RESULTS: STEEL PROFILE		Reference	ABCPMX	Dev.
Cross-sectional area : $A =$	(cm ²)	202,9	202,9	
Shear area : $A_v =$	(cm ²)	39,04	39,04	
Second moment of area (y-y axis) : $I_y =$	(cm ⁴)	23 274	23 274	
Second moment of area (z-z axis) : $I_z =$	(cm ⁴)			
Mass by length unit : $m_{lin} =$	(kg/ml)	159,31	159,3	
Mass : $m =$	(kg)	1274,5	1274	
Nominal yield strength : $f_y =$	(MPa)	345	345	
Axial resistance : $N_{t,Rd} =$	(kN)	7001	7001	
Shear resistance : $V_{pl,Rd} =$	(kN)	777,6	778	
Plastic moment resistant : $M_{pl,Rd} =$	(kN.m)	696	696	
Location of the plastic neutral axis: $z_{ANP} =$	(mm)	10,88	10,88	
Elastic moment resistant : $M_{el,Rd} =$	(kN.m)	572	572	

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	106	Rév. : D

CASE SFS02 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	53,93	53,93	
Bending moment : $M(G_1) =$ (kN.m)	107,9	107,9	
Shear force : $V(Q_1\#1) =$ (kN)	120	120	
Bending moment : $M(Q_1\#1) =$ (kN.m)	240	240	
Shear force : $V_{Ed,ULS} =$ (kN)	252,8	252,8	
Bending moment : $M_{Ed,ULS} =$ (kN.m)	505,6	505,6	

CASE SFS02 RESULTS: ULS IN THE CENTRAL SPAN	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_{M,ULS} =$	0,736	0,736	
Resistance to shear force : $\Gamma_{V,ULS} =$	0,325	0,325	
Resistance to local bending : $\Gamma_{MV,ULS} =$	0,240	0,240	

CASE SFS02 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	14,7	14,7	
Deflection : $\delta(Q_1)$ (mm)	32,7	32,7	
Deflection : $\delta(SLS)$ (mm)	47,5	47,4	

8.4 CASE SFS03: Steel beam with IFB-B steel section

8.4.1 Description

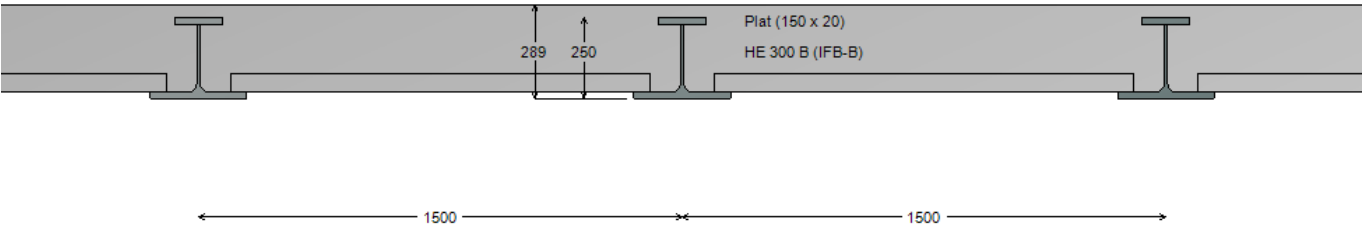


Figure 29 : Transversal view

8.4.2 Parameters

Definition of the beam:

Definition de la poutre

Portées

Travée principale

L = 7,00 m

Entraxes

☒ Poutre intermédiaire

☐ Poutre de rive

Entraxes

d₁ = 1,50 m

d₂ = 1,50 m

Annuler OK

Definition of the profile:

Profile: HEB 300

Steel grade: S355 J0

Steel plate

Dimensions: 150x20

Steel grade: S355 J0

	Project: ABCPMX-II Software - DAL-22-003						
	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	109	Rév. : D

8.4.3 Verification of results

CASE SFS03 RESULTS: STEEL PROFILE		Reference	ABCPMX	Dev.
Cross-sectional area : $A =$	(cm ²)	113,34	113,3	
Shear area : $A_v =$	(cm ²)	32,51	32,51	
Second moment of area (y-y axis) : $I_y =$	(cm ⁴)	11718	11718	
Second moment of area (z-z axis) : $I_z =$	(cm ⁴)	4843	4840	
Mass by length unit : $m_{lin} =$	(kg/ml)	88,97	89,0	
Mass : $m =$	(kg)	622,8	622,8	
Nominal yield strength : $f_y =$	(MPa)	345	345	
Axial resistance : $N_{t,Rd} =$	(kN)	3910	3910	
Shear resistance : $V_{pl,Rd} =$	(kN)	647,6	648	
Plastic moment resistant : $M_{pl,Rd} =$	(kN.m)	332,5	333	
Location of the plastic neutral axis: $z_{ANP} =$	(mm)	-0,110	-0,117	
Elastic moment resistant : $M_{el,Rd} =$	(kN.m)	262	260	

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	110	Rév. : D

CASE SFS03 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	38,46	38,46	
Bending moment : $M(G_1) =$ (kN.m)	67,3	67,3	
Shear force : $V(Q_1) =$ (kN)	52,5	52,5	
Bending moment : $M(Q_1) =$ (kN.m)	91,9	91,9	
Shear force : $V_{Ed,ULS} =$ (kN)	130,7	130,7	
Bending moment : $M_{Ed,ULS} =$ (kN.m)	228,7	228,7	

CASE SFS03 RESULTS: ULS IN THE CENTRAL SPAN	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_{M,ULS} =$	0,688	0,688	
Resistance to shear force : $\Gamma_{V,ULS} =$	0,202	0,202	
Resistance to local bending : $\Gamma_{MV,ULS} =$	0,0506	0,051	

CASE SFS03 RESULTS: SLS	Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$ (mm)	13,7	13,7	
Deflection : $\delta(Q_1)$ (mm)	19,1	19,1	
Deflection : $\delta(SLS)$ (mm)	32,8	32,8	

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	111	Rév. : D

8.5 CASE SFS04: Steel beam with SAB steel section

8.5.1 Description

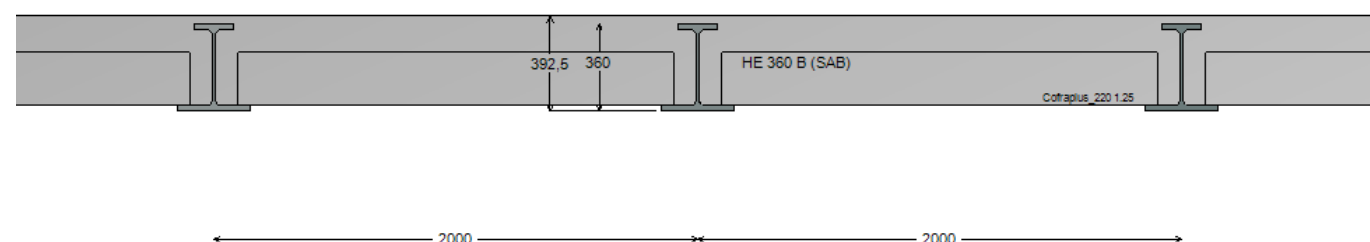


Figure 30 : Transversal view

8.5.2 Parameters

Definition of the beam:

Definition of the profile:

Profile: HEB 360
Steel grade: S420 EC3
Upper flange width: 160

Definition of loads:

Définition de la dalle

Général Armatures **Bac**

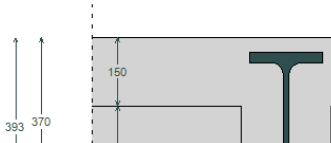
Bac acier

Nom Cofraplus_220 1.25



Hauteur $h_p = 220.0$ mm

Modifier le bac acier



370 393

Annuler OK

live load Q_1 : surface load: 10 kN/m^2

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	113	Rév. : D

8.5.3 Verification of results

CASE SFS03 RESULTS: STEEL PROFILE	Reference	ABCPMX	Dev.
Cross-sectional area : $A =$ (cm ²)	149,1	149,1	
Shear area : $A_v =$ (cm ²)	60,59	60,6	
Second moment of area (y-y axis) : $I_y =$ (cm ⁴)	32315	32315	
Second moment of area (z-z axis) : $I_z =$ (cm ⁴)	5842	5836	
Mass by length unit : $m_{lin} =$ (kg/ml)	117,1	117,1	
Mass : $m =$ (kg)	995,1	995,1	
Nominal yield strength : $f_y =$ (MPa)	420	420	
Axial resistance : $N_{t,Rd} =$ (kN)	6264	6264	
Shear resistance : $V_{pl,Rd} =$ (kN)	1469	1469	
Plastic moment resistant : $M_{pl,Rd} =$ (kN.m)	820,3	820	
Location of the plastic neutral axis: $z_{ANP} =$ (mm)	31,5	31,5	
Elastic moment resistant : $M_{el,Rd} =$ (kN.m)	629	629	

CASE SFS03 RESULTS: ANALYSIS	Reference	ABCPMX	Dev.
Shear force : $V(G_1) =$ (kN)	50,05	50,05	
Bending moment : $M(G_1) =$ (kN.m)	112,6	112,6	
Shear force : $V(Q) =$ (kN)	90	90	
Bending moment : $M(Q) =$ (kN.m)	202,5	202,5	
Shear force : $V_{Ed,ULS} =$ (kN)	202,7	202	
Bending moment : $M_{Ed,ULS} =$ (kN.m)	455,8	455,8	

CASE SFS03 RESULTS: ULS IN THE CENTRAL SPAN	Reference	ABCPMX	Dev.
Resistance to bending moment : $\Gamma_{M,ULS} =$	0,557	0,557	
Resistance to shear force : $\Gamma_{V,ULS} =$	0,138	0,138	
Resistance to local bending : $\Gamma_{MV,ULS} =$	0,044	0,044	

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	Product reference: DRV/PMX/MV/001						
	Date:	15/05/2025	Authors:	POM+GUD	Page:	114	Rév. : D

CASE SFS03 RESULTS: SLS		Reference	ABCPMX	Dev.
Deflection : $\delta(G_1)$	(mm)	14,0	14,0	
Deflection : $\delta(Q_1)$	(mm)	25,2	25,2	
Deflection : $\delta(SLS)$	(mm)	39,2	39,2	